

Command filtered adaptive bipartite synchronization for heterogeneous switched multi-agent systems with state delays and input nonlinearity

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ABSTRACT

In this paper, we investigate a distributed protocol design to achieve bipartite synchronization for a group of heterogeneous switched multi-agent systems (MASs) with state delays and input nonlinearity in each agent's dynamics. To achieve this aim, the neural network-based approximation method is combined with an adaptive control method to stabilize each agent with uncertain dynamics. A two-layered compensation mechanism is also developed to attenuate the effect of input nonlinearity, consisting of two filters: one to address the input dead-zone and one to reduce the dynamic surface errors. The distributed command filters are introduced for the cascaded structure of each agent's dynamics to eradicate recursive virtual control differentiation while diminishing the filtered error for each agent. Since it is challenging to address multiple distinct state delays for a group of MASs with asynchronous switching, the Lyapunov-Krasovskii functional design is extended and combined with the average dwell time (ADT) method. Bipartite synchronization of a group of switched nonlinear agents is ensured with the presented design and the boundedness of closed-loop signals is discussed through theoretical analysis. Finally, a simulation example is conducted to validate the distributed protocol design for bipartite synchronization.

1. Introduction

Multi-agent systems (MASs) have attracted extensive attention in recent years because they offer high flexibility, scalability, and resilience in cooperative tasks. Cooperative synchronization protocols [1] have been developed in various forms, such as leaderless consensus [2–4], leader-following synchronization [5–7], and containment control [8–10]. These protocols enable groups of agents to coordinate their behaviour using only local information exchange. In parallel, function approximation techniques, including neural networks and related intelligent adaptive control methods [11–13], have been successfully applied to handle complex nonlinear dynamics such as time-delay effects [14,15] and switching behaviors [16–18] in individual agents. Input dead-zone nonlinearities are another important feature in many engineering systems, such as servo mechanisms [19,20] and mechanical systems [21,22]. If such nonlinearities are not explicitly compensated, they may deteriorate tracking performance or even destabilize the closed-loop system. Consequently, it is necessary to design distributed protocols for MASs that explicitly take into account input dead-zone effects while coping with heterogeneous nonlinear dynamics and switching behaviours.

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Related Work

In a group of MASs, the distributed interactions between agents are modeled using graph theory, with the nodes and edges denoting each agent and the communication link between two distinct agents. For cooperative protocol designs, these distributed interactions are mainly designed based on the communication topology captured by the unsigned graphs [23,24], which has been widely applied to areas including multi-robot systems and multi-spacecraft systems [25]. In contrast, competitive interactions between a group of MASs require different communication topologies. For example, in one robotic soccer game, the teammates in a group and the opponents in another group constitute zero-sum objectives, where the communication topology is represented as a signed graph to consider the antagonistic interactions. Since bipartite consensus was introduced, it has been widely investigated for MASs under different circumstances [26,27], including actuation saturation [28] and a leader with autonomous dynamics [29]. However, the agent dynamics presented in the literature [28,29] are in a linear form, which cannot be directly extended to heterogeneous MASs with nonlinear agent dynamics. In this paper, the bipartite synchronization protocol design is considered for a group of heterogeneous nonlinear MASs, with each agent possessing multiple switched dynamics and time-varying delays in its own dynamics.

In many engineering applications, the plant dynamics can operate in several distinct modes, which cannot be adequately captured by a single model. Typical examples include robotic manipulators that switch between on-load and off-load modes, or groups of agents that change their operating conditions over time [30]. Common Lyapunov function (CLF) methods have been developed for nonlinear strict-feedback and non-triangular systems [31,32], but constructing a CLF can be difficult under certain circumstances, such as the common Lyapunov-Krasovskii functional design for switched time-delay systems. Multiple Lyapunov function (MLF) techniques therefore offer useful alternatives, where stability is guaranteed by jointly designing mode-dependent Lyapunov functions and constraining the switching frequency [33]. Moreover, instead of investigating the behavior of Lyapunov candidates at switching instants, the concept of average dwell time (ADT) approaches is to only take the switching frequency into account [34]. This approach has been successfully applied to switched systems with dynamical uncertainty [35], Takagi-Sugeno fuzzy systems [36], and circuit systems [37]. In this paper, we extend the ADT-based method for bipartite synchronization of a group of MASs with asynchronous switching.

In real-world applications, the plant and controller are usually remotely located, meaning time delay effects cannot be neglected [38]. This communication delay was considered for tracking consistency regulation of a group of perturbed linear MASs with an aperiodic sampling scheme to reduce the information exchange frequency [39]. There exist other types of time delays in the dynamics of each agent, including input delays [40,41] and state delays [42], which generally exist in applications [43]. In contrast to constant-type delays [44], time-varying delays are more general and widely exist in engineering systems [45]. The containment control problem is studied for a group of unmodeled dynamical MASs with time-varying state delays [46]. In addition, distributed finite-time consensus protocols have been designed for MASs with second-order agent dynamics using an event-triggered sampling mechanism [47]. In this study, we investigated the bipartite synchronization problem for heterogeneous MASs with time-varying full-state delays.

Contributions. The contributions of this paper are twofold.

1. Compared with existing bipartite protocol designs for linear MASs [28,29], this paper investigates nonlinear switched MASs with time-varying state delays, where the delay-dependent couplings among agents significantly complicate the design and analysis of bipartite protocols. In particular, the joint effect of asynchronous switching and full time-varying state delays on bipartite synchronization for a group of heterogeneous agents is explicitly addressed via a novel Lyapunov-Krasovskii functional that captures both the switching behavior and the temporal variation of delays in a unified framework.
2. In contrast to centralized command-filtering schemes [48,49], the proposed command filters are implemented in a fully distributed fashion. By embedding the distributed command filters into the bipartite synchronization protocol, each agent can attenuate dynamic surface errors and avoid repeated differentiation while preserving a purely local information structure, which leads to improved synchronization performance under asynchronous switching and time-varying state delays.

Structure. The remainder of this paper is structured as follows. Background knowledge about the communication topology of a group of competitive MASs and the bipartite synchronization problem are formulated in Section 2. The distributed bipartite synchronization protocol design for each agent is provided in Section 3 and a Lyapunov-based stability analysis for the overall design is presented in Section 4. A simulation example is implemented in Section 5. Finally, the concluding remarks are presented in Section 6.

2. Preliminaries and problem formulation

2.1. Mathematical preliminaries

2.1.1. Radial basis function neural networks

Radial basis function neural networks (RBFNNs) are powerful tools for addressing function approximation problems. To approximate a continuous function $P(X) : \Omega_X \rightarrow \mathbb{R}$, the RBFNNs $\hat{P}(X)$ under consideration adopt the form of $\hat{P}(X) = W^T \phi(X)$, where $W \in \mathbb{R}^m$ and $X \in \Omega_X$ are the weight and input vectors, respectively. Let $\phi(X) = [\phi_1(X), \phi_2(X), \dots, \phi_m(X)]^T \in \mathbb{R}^m$ with $\phi_i(X) = \exp\left(-\frac{\|X - \mu_i\|^T \|X - \mu_i\|}{2\sigma^2}\right)$ be the basis function, while μ_i and σ are the center and width, respectively, for $i = 1, 2, \dots, m$.

Lemma 1. [50] Consider the RBFNN $\hat{P}(X) = W^T \phi(X)$ defined on $\Omega_X \subseteq \mathbb{R}^p$ with the m -dimensional weight $W \in \mathbb{R}^m$, the radial basis function $\phi(X) = [\phi_1(X), \dots, \phi_m(X)]^T$, and $\phi_i(X) = \exp\left(-\frac{[X-\mu_i]^T[X-\mu_i]}{2\sigma^2}\right)$ with μ_i and σ being the center and width for $i = 1, \dots, m$, respectively. Denote $\bar{\mu} = \frac{1}{2} \min_{i \neq j} \|\mu_i - \mu_j\|$. Then, one has $\|\phi(X)\| \leq \bar{\phi}, \forall X \in \Omega_X$ with $\bar{\phi}$ being a constant independent of the X and m .

Lemma 2. [51] On a compact set Ω_X , given $P(\cdot) : \Omega_X \rightarrow \mathbb{R}$, $\varepsilon > 0$ there exists an RBFNN $\hat{P}(X) = W^T \phi(X)$ with sufficiently many basis number m , such that $\sup_{X \in \Omega_X} |P(X) - W^T \phi(X)| \leq \varepsilon$. In addition, the optimal weight is $W^* = \arg \min_{W \in \mathbb{R}^m} \left\{ \sup_{X \in \Omega_X} |P(X) - W^T \phi(X)| \right\}$, with the residual $\varepsilon^*(X) = P(X) - W^{*T} \phi(X)$ satisfying $\sup_{X \in \Omega_X} |\varepsilon^*(X)| \leq \varepsilon$.

2.1.2. Graph theory

In a group of MASs, the communication topology among all followers (or agents) is denoted as a graph $\mathcal{G}_N$ (or $\mathcal{G}_{N+1}$) with node set $\mathcal{F} = \{1, \dots, N\}$ (or $\mathcal{V} = \{0, 1, \dots, N\}$) and edge set $\mathcal{E}_N \subseteq \mathcal{F} \times \mathcal{F}$ (or $\mathcal{E}_{N+1} \subseteq \mathcal{V} \times \mathcal{V}$), where the leader and the follower set are denoted as 0 and $\mathcal{F}$, respectively. $\mathcal{G}_N$ is considered a signed graph with the adjacency matrix $\mathcal{A} = [a_{i,j}] \in \mathbb{R}^{N \times N}$, where element $a_{i,j}$ could be either positive or negative. Denote $\mathcal{N}_i = \{j | a_{i,j} \neq 0\}$ as the neighborhood set of the i -th follower. Then, signed graph $\mathcal{G}_N$ is structurally balanced if bipartition $\mathcal{F}_1$ and $\mathcal{F}_2$ satisfy $\mathcal{F}_1 \cup \mathcal{F}_2 = \mathcal{F}$, $\mathcal{F}_1 \cap \mathcal{F}_2 = \emptyset$ [29]. Both the i -th and j -th followers belong to $\mathcal{F}_c$ with $c = 1, 2$ if $(j, i) \in \mathcal{E}_N$ with $a_{i,j} > 0$. In contrast, the i -th follower belongs to $\mathcal{F}_c$ and the j -th follower belongs to $\mathcal{F}_{3-c}$ with $c = 1, 2$ if $(j, i) \in \mathcal{E}_N$ with $a_{i,j} < 0$. The interaction between the leader and the i -th follower is captured by the pinning gain b_i , where $b_i > 0$ if the i -th follower is connected to the leader with $i \in \mathcal{F}_1$, and $b_i < 0$ if the i -th follower is connected to the leader with $i \in \mathcal{F}_2$, and $b_i = 0$, otherwise.

2.2. System model

In this paper, the bipartite objective is captured by the leader, which is denoted as x_0 and assumed to be sufficiently smooth and differentiable. In addition, the leader information is only available for the i -th follower if $b_i \neq 0$.

The i -th follower is considered with the following dynamics

$$\begin{aligned} \dot{x}_{i,l} &= f_{i,l}^{\sigma_i(t)}(\bar{x}_{i,l}) + g_{i,l}^{\sigma_i(t)}(\bar{x}_{i,l}(t - \tau_{i,l}^{\sigma_i(t)})) + x_{i,l+1}, \\ \dot{x}_{i,n_i} &= f_{i,n_i}^{\sigma_i(t)}(x_i) + g_{i,n_i}^{\sigma_i(t)}(x_i(t - \tau_{i,n_i}^{\sigma_i(t)})) + \Psi_i(u_i), \\ y_i &= x_{i,1}, \end{aligned} \quad (1)$$

for $l = 1, \dots, n_i - 1$, where $\bar{x}_{i,l} = [x_{i,1} \dots x_{i,l}]^T \in \mathbb{R}^l$ with $x_i = \bar{x}_{i,n_i} \in \mathbb{R}^{n_i}$ is the state, $y_i \in \mathbb{R}$ is the output, and $u_i \in \mathbb{R}$ is the input. In this paper, the switching of the followers is *asynchronous*. That is, each follower's dynamics consists of multiple switching modes with index set $\mathcal{Q}_i = \{1, \dots, Q_i\}$, where Q_i is the number of modes for the i -th follower. The piecewise continuous switching signal for the i -th follower is denoted as $\sigma_i : [t_0, +\infty) \rightarrow \mathcal{Q}_i$. The switching signal $\sigma_i(\cdot)$ determines the switching sequence for all the subsystems of the i -th follower, i.e., $\left\{ (q_{i,0}, t_{i,0}, \tau_{i,l}^{q_{i,0}}), \dots, (q_{i,k}, t_{i,k}, \tau_{i,l}^{q_{i,k}}), \dots \right\}$. In addition, $\sigma_i(t) = q_{i,k} \in \mathcal{Q}_i$ for $t \in [t_{i,k}, t_{i,k+1})$ denotes that the $q_{i,k}$ -th mode is active for the i -th follower during $[t_{i,k}, t_{i,k+1})$, where $\tau_{i,l}^{q_{i,k}}$ is the time-varying state delay. For each $q_{i,k} \in \mathcal{Q}_i$, the functions $f_{i,l}^{q_{i,k}}(\cdot) : \mathbb{R}^l \rightarrow \mathbb{R}$ and $g_{i,l}^{q_{i,k}}(\cdot) : \mathbb{R}^l \rightarrow \mathbb{R}$ are unknown but continuous, such that the solution to system (1) exists and is unique. The follower states do not jump at switching instants. In the agent dynamics, the input nonlinearity of interest is the dead-zone mapping, denoted as $\Psi_i(u_i)$ with

$$\Psi_i(u_i) = \begin{cases} \Psi_{i,L}(u_i), & u_i \leq -u_{i,L}, \\ 0, & -u_{i,L} < u_i < u_{i,R}, \\ \Psi_{i,R}(u_i), & u_i \geq u_{i,R}, \end{cases} \quad (2)$$

where $\Psi_{i,L}(\cdot) : \mathbb{R} \rightarrow \mathbb{R}$ and $\Psi_{i,R}(\cdot) : \mathbb{R} \rightarrow \mathbb{R}$ are dead-zone mappings with $\Psi_{i,L}(-u_{i,L}) = \Psi_{i,R}(u_{i,R}) = 0$, and

$$\begin{aligned} 0 < \underline{\Psi}_{i,L} \leq \Psi'_{i,L}(u_i) \leq \bar{\Psi}_{i,L}, \quad u_i \in (-\infty, -u_{i,L}], \\ 0 < \underline{\Psi}_{i,R} \leq \Psi'_{i,R}(u_i) \leq \bar{\Psi}_{i,R}, \quad u_i \in [u_{i,R}, +\infty). \end{aligned} \quad (3)$$

According to [21], the dead-zone mapping $\Psi_i(\cdot)$ satisfies $\Psi_i(u_i) \leq d_i \|u_i\| + c_i$ for all $u_i \in \mathbb{R}$ with positive constants $d_i = \bar{\Psi}_{i,L} + \bar{\Psi}_{i,R}$ and $c_i = (\bar{\Psi}_{i,L} + \bar{\Psi}_{i,R}) \max\{u_{i,L}, u_{i,R}\}$.

Definition 1. [34] The switching signal $\sigma(t)$ has an average dwell time (ADT) $\mathcal{T}_a$ if there exists a chatter bound $\bar{N}$ such that $N_\sigma(t_0, t_f) \leq \bar{N} + \frac{t_f - t_0}{\mathcal{T}_a}$, where $N_\sigma(t_0, t_f)$ denotes the discontinuous point number of $\sigma(t)$ on the interval (t_0, t_f) .

Assumption 1. The signed graph $\mathcal{G}_{N+1}$ is connected and its subgraph $(\mathcal{G}_N)$ is structurally balanced.

Assumption 2. The switching signal $\sigma_i(t)$ has an ADT $\mathcal{T}_{a,i}$.

Assumption 3. For the time-varying delay, there exist constants $\bar{\tau}_{d,i}, \bar{\tau}_i \in \mathbb{R}^+$ such that $\bar{\tau}_{i,l}^{\sigma_i(t)}(t) \leq \bar{\tau}_{d,i} < 1$ and $0 \leq \tau_{i,l}^{\sigma_i(t)}(t) \leq \bar{\tau}_i$, for all $i \in \mathcal{F}$, $l \in \{1, \dots, n_i\}$ and $q_{i,k} \in \mathcal{Q}_i$.

Table 1
Main notation used in the controller and error definitions.

Symbol	Description
$x_{i,l}$	l -th state of the i -th follower
u_i	Control input of the i -th follower
$\Psi_i(u_i)$	Input dead-zone nonlinearity of the i -th follower
$s_{i,l}$	Virtual control signal at layer l
$\alpha_{i,l}$	DSC-filtered version of $s_{i,l}$
$\tilde{\alpha}_{i,l}$	DSC filter error, $\tilde{\alpha}_{i,l} = \alpha_{i,l} - s_{i,l}$
$\lambda_{i,l}$	High-order compensation state for input dead-zone
$\beta_{i,l}$	Compensation signal for DSC filter errors
$\xi_{i,l}$	Tracking error, $\xi_{i,l} = x_{i,l} - \alpha_{i,l-1}$
$\delta_{i,l}$	Compensated tracking error, $\delta_{i,l} = \xi_{i,l} - \lambda_{i,l}$
$v_{i,l}$	Final error signal, $v_{i,l} = \delta_{i,l} - \beta_{i,l}$
$\hat{\Theta}_{i,l}$	Adaptive estimate of the (squared) NN weight bound
$\tilde{\Theta}_{i,l}$	Estimation error, $\tilde{\Theta}_{i,l} = \Theta_{i,l} - \hat{\Theta}_{i,l}$

Remark 1. Structural balance of the signed graph in [Assumption 1](#) is realistic in many cooperative-competitive MASs, such as two-team coordination, clustering of agents into two antagonistic groups, or friend-enemy networks. In practice, this partition can often be designed offline by assigning agents to two coalitions based on their objectives or interaction patterns. Once the two groups are specified, the corresponding signed graph satisfies the structural balance condition, which is a standard assumption in bipartite consensus and synchronization studies.

Remark 2. The ADT condition in [Assumption 2](#) is a classical requirement in switched-systems analysis. It prevents arbitrarily fast switching, which could destabilize the overall system even if each mode is stable. Technically, the ADT guarantees that the negative decrease of the Lyapunov-Krasovskii functionals within each mode dominates the possible increases at switching instants. In practical MASs, such a bound is consistent with physical switching limitations or supervisory constraints on mode transitions.

Remark 3. The bounds on the time-varying delays and their derivatives are standard in the literature on nonlinear time-delay systems. The condition $\dot{\tau}_{i,l}^{(t)} < 1$ ensures that the Lyapunov-Krasovskii functionals remain positive and that the integral terms associated with the delayed states can be properly bounded when computing the time derivative. This will be discussed in the stability analysis in [Section 4](#). This assumption is mild, since communication or processing delays in practice typically evolve more slowly than the physical time scale of the plant dynamics.

Remark 4. To verify [Assumption 3](#), for example, the time-varying state delays are chosen as $\tau_{i,l}^{\sigma_i(t)}(t) = \frac{\bar{\tau}_i}{2} [1 + \sin(t)]$ for the corresponding modes, where $\bar{\tau}_i > 0$ is a prescribed upper bound. It is straightforward to verify that $0 \leq \tau_{i,l}^{\sigma_i(t)}(t) \leq \bar{\tau}_i$ and $\dot{\tau}_{i,l}^{\sigma_i(t)}(t) = \frac{\bar{\tau}_i}{2} \cos(t)$, so that $|\dot{\tau}_{i,l}^{\sigma_i(t)}(t)| \leq \bar{\tau}_i/2 < 1$ if $\bar{\tau}_i < 2$. Hence, [Assumption 3](#) is satisfied with $\tau_{d,i} = \bar{\tau}_i/2$ in this example.

2.3. Bipartite synchronization objective

With the communication topology and the asynchronous switching, the local error dynamics of agent i therefore involve both its own mode $\sigma_i(t)$ and its neighbors $j \in \mathcal{N}$, which can be described as

$$\delta_{i,1} = \sum_{j \in \mathcal{N}_i} |a_{i,j}| [x_{i,1} - \text{sign}(a_{i,j})x_{j,1}] + |b_i| [x_{i,1} - \text{sign}(b_i)x_0], \quad (4)$$

In subsequent discussions, term $\delta_{i,1}$ is referred to as the bipartite synchronization error.

Definition 2. [52] MASs achieve bipartite synchronization if the output y_i of each follower satisfies $y_i = x_0$ for $i \in \mathcal{F}_1$, and $y_i = -x_0$ for $i \in \mathcal{F}_2$, where x_0 denotes the leader state.

In subsequent discussions, $\delta_{i,1}$ is referred to as the bipartite synchronization error. According to [52], bipartite synchronization can be achieved in the *ideal* case if $\xi_{i,1} \rightarrow 0$ as $t \rightarrow +\infty$, where $\xi_{i,1}$ denotes the local tracking error with respect to the signed leader-follower interaction.

Under the uncertainties (e.g. time-varying state delays, dead-zone input nonlinearities, approximation residuals, etc.) and asynchronous switching. We aim to design a distributed protocol such that the MASs achieve *practical* bipartite synchronization with the error $\delta_{i,1}$ being driven to a small neighborhood of the origin while all closed-loop signals remain bounded.

3. Distributed protocol design

In this section, a two-layer compensation mechanism is presented to enhance the overall collective behavior of the MASs to achieve the bipartite synchronization target. Based on a two-layer filter design, a distributed bipartite synchronization protocol design for a group of heterogeneous MASs with asynchronous switching and time-varying state delays is presented. To begin with, [Table 1](#) summarizes the main variables and signals of this section for notation consistency and readability.

3.1. Distributed two-layer compensation mechanism

To reduce computational complexity, the DSC filter is introduced as

$$\omega_{i,l} \dot{\alpha}_{i,l} = s_{i,l} - \alpha_{i,l}, \quad 1 \leq l \leq n_i - 1 \quad (5)$$

with $\alpha_{i,l}(0) = s_{i,l}(0)$ and $\omega_{i,l} > 0$.

Next, we develop a two-layer compensation mechanism for the heterogeneous MASs. The first-layer command filter is introduced to avoid repeated differentiation of the auxiliary signals and eliminate the impact of interactions between agents on the bipartite synchronization target. To compensate for the filtered error, the compensating signals $\{\beta_{i,l}\}_{l=1}^{n_i}$ are designed as

$$\dot{\beta}_{i,1} = -\Lambda_{i,1}\beta_{i,1} + \bar{\Lambda}_{i,2}\beta_{i,2} + o_i \tilde{\alpha}_{i,1}, \quad (6)$$

$$\dot{\beta}_{i,l} = -\underline{\Lambda}_{i,l-1}\beta_{i,l-1} - \Lambda_{i,l}\beta_{i,l} + \bar{\Lambda}_{i,l+1}\beta_{i,l+1} + \tilde{\alpha}_{i,l}, \quad (7)$$

$$\dot{\beta}_{i,n_i} = -\underline{\Lambda}_{i,n_i-1}\beta_{i,n_i-1} - \Lambda_{i,n_i}\beta_{i,n_i}, \quad (8)$$

with $2 \leq l \leq n_i - 1$ and initial conditions $\beta_{i,1}(0) = \beta_{i,l}(0) = \beta_{i,n_i}(0) = 0$. The design parameters $\{\underline{\Lambda}_{i,l}\}_{l=1}^{n_i-1}$ and $\{\bar{\Lambda}_{i,l}\}_{l=2}^{n_i}$ are selected as

$$\begin{aligned} \bar{\Lambda}_{i,2} &= o_i, \quad \bar{\Lambda}_{i,l} = 1, \quad l = 3, \dots, n_i, \\ \underline{\Lambda}_{i,1} &= o_i, \quad \underline{\Lambda}_{i,l} = 1, \quad l = 2, \dots, n_i - 1. \end{aligned} \quad (9)$$

The second-layer high-order filter design aims to make amends for the input dead-zone of each follower. To address the input dead-zone (2), we design

$$\begin{aligned} \dot{\lambda}_{i,l} &= -\rho_{i,l}\lambda_{i,l} + \lambda_{i,l+1}, \quad l = 1, \dots, n_i - 1, \\ \dot{\lambda}_{i,n_i} &= -\bar{\rho}_{i,n_i}\rho_{i,n_i}\lambda_{i,n_i} + \Psi_i(u_i) - u_i. \end{aligned} \quad (10)$$

for $\lambda_{i,l}(0) = \lambda_{i,n_i}(0) = 0$, with design parameters $\rho_{i,l} > 0$ and $\bar{\rho}_{i,n_i} \geq 1 + \max\{c_{i,4}, c_{i,6}\}$.

Remark 5. The command filters in (6)–(8) are distributed in the sense that each agent computes its filtered and compensating signals using only its own state and neighbor information. This distributed implementation mitigates the growth of communication and computation burden in large-scale networks, and allows the proposed scheme to maintain improved synchronization performance without relying on a centralized coordinator.

3.2. Distributed bipartite synchronization protocols design

For each follower, the tracking errors $\delta_{i,l}$, filtered errors $\tilde{\alpha}_{i,l}$, and compensated errors $v_{i,l}$ are defined as

$$\begin{aligned} \xi_{i,l} &= x_{i,l} - \alpha_{i,l-1}, \quad l = 2, \dots, n_i, \\ \delta_{i,l} &= \xi_{i,l} - \lambda_{i,l}, \quad l = 2, \dots, n_i, \\ \tilde{\alpha}_{i,l} &= \alpha_{i,l} - s_{i,l}, \quad l = 1, \dots, n_i - 1, \\ v_{i,l} &= \delta_{i,l} - \beta_{i,l}, \quad l = 1, \dots, n_i, \end{aligned} \quad (11)$$

where $s_{i,l}$ is the virtual control and $\beta_{i,l}$ is the compensating signal.

3.2.1. Step 1

Consider the agent dynamics as in (1) and (4), differentiating $\delta_{i,1}$ yields

$$\begin{aligned} \dot{\delta}_{i,1} &= o_i [f_{i,1}^{\sigma_i(t)}(x_{i,1}) + g_{i,1}^{\sigma_i(t)}(x_{i,1}(t - \tau_{i,1}^{\sigma_i(t)})) + s_{i,1} + \tilde{\alpha}_{i,1} + \delta_{i,2} + \lambda_{i,2}] - b_i \dot{x}_0 \\ &\quad - \sum_{j \in \mathcal{N}_i} a_{i,j} [f_{j,1}^{\sigma_j(t)}(x_{j,1}) + x_{j,2} + g_{j,1}^{\sigma_j(t)}(x_{j,1}(t - \tau_{j,1}^{\sigma_j(t)}))] \end{aligned} \quad (12)$$

with $o_i = |b_i| + \sum_{j \in \mathcal{N}_i} |a_{i,j}|$.

Remark 6. Compared with the centralized command-filtered designs in [45,46], the proposed two-layer compensation mechanism exhibits both structural similarities and important differences. Similar to [48,49], the DSC filter in (5) is introduced to avoid recursive differentiation of the virtual controls $s_{i,l}$ in the strict-feedback structure, and the compensating signal $\beta_{i,l}$ in (6)–(8) is designed to attenuate the resulting filtered error. However, in contrast to the centralized setting of [45,46], our filter design is distributed and explicitly coupled to the signed graph through the bipartite error $\delta_{i,1}$ and the aggregate gain o_i in (12). Moreover, a second-layer high-order filter $\lambda_{i,l}$ is introduced in (10) to compensate the input dead-zone nonlinearity $\Psi_i(u_i)$, which is not addressed in [48,49].

Define $X_{i,1} = [x_{i,1}, x_{j,1}, v_{i,1}]^T$ and

$$F_{i,1}^{\sigma_i(t)}(X_{i,1}) = o_i f_{i,1}^{\sigma_i(t)}(x_{i,1}) - \sum_{j \in \mathcal{N}_i} a_{i,j} f_{j,1}^{\sigma_j(t)}(x_{j,1}),$$

$$G_{i,1}^{\sigma_i(t)} = e^{\varpi_i \bar{\tau}_i} \left[g_{i,1}^{\sigma_i(t)}(x_{i,1}) \right]^2 + \sum_{j \in \mathcal{N}_i} e^{\varpi_j \bar{\tau}_j} \left[g_{j,1}^{\sigma_j(t)}(x_{j,1}) \right]^2,$$

$$H_{i,1}^{\sigma_i(t)}(X_{i,1}) = F_{i,1}^{\sigma_i(t)}(X_{i,1}) + 2 \tanh^2 \left(\frac{v_{i,1}}{\chi_{i,1}} \right) \frac{G_{i,1}^{\sigma_i(t)}}{v_{i,1}} + \frac{1}{4} \left[\frac{o_i^2}{(1 - \bar{\tau}_{d,i})} + \sum_{j \in \mathcal{N}_i} \frac{1}{(1 - \bar{\tau}_{d,j})} \right] v_{i,1}.$$

To address the unknown functions $f_{i,1}^{\sigma_i(t)}(\cdot)$ and $g_{i,1}^{\sigma_i(t)}(\cdot)$ in $F_{i,1}^{\sigma_i(t)}(X_{i,1})$, $G_{i,1}^{\sigma_i(t)}(X_{i,1})$, and $H_{i,1}^{\sigma_i(t)}(X_{i,1})$ defined previously, based on the universal approximation theorem [53] and given an accuracy level of $\varepsilon_{i,1} > 0$, there exists an RBFNN to represent the function $H_{i,1}^{q_{i,k}}$ such that

$$H_{i,1}^{q_{i,k}}(\cdot) = \left(\theta_{i,1}^{q_{i,k}} \right)^T \phi_{i,1}(\cdot) + \varepsilon_{i,1}^{q_{i,k}}(\cdot), \forall q_{i,k} \in \mathcal{Q}_i \quad (13)$$

with the residual satisfying $|\varepsilon_{i,1}^{q_{i,k}}(\cdot)| \leq \frac{\varepsilon_{i,1}^{q_{i,k}}}{\eta_{i,1}^X}$ on $\Omega_{i,1}^X$, where $\theta_{i,1}^{q_{i,k}} = \arg \min_{\theta \in \Omega_{i,1}^{\theta}} \sup_{X \in \Omega_{i,1}^X} \|H_{i,1}^{q_{i,k}}(X) - \theta^T \phi_{i,1}(X)\|$, and $\phi_{i,1}(X_{i,1}) = [\phi_{i,1,1}(X_{i,1}), \dots, \phi_{i,1,N_{i,1}}(X_{i,1})]^T$ is the $N_{i,1}$ -dimensional radial basis function. It should be noted that both the ideal weight $\theta_{i,1}^{q_{i,k}}$ exist, although they are unknown for all $q_{i,k} \in \mathcal{Q}_i$. Then, $\Theta_{i,1} = \max_{q_{i,k} \in \mathcal{Q}_i} \|\theta_{i,1}^{q_{i,k}}\|^2$ is also unavailable for controller design. To address this issue, we consider

$$\dot{\hat{\Theta}}_{i,1} = -\Gamma_{i,1} \hat{\Theta}_{i,1} + \frac{1}{2\eta_{i,1}^2} v_{i,1}^T \phi_{i,1}^T \phi_{i,1}, \quad \hat{\Theta}_{i,1}(0) \geq 0, \quad (14)$$

with learning error $\bar{\Theta}_{i,1} = \Theta_{i,1} - \hat{\Theta}_{i,1}$ and design parameters $\Gamma_{i,1}$, $\eta_{i,1}$.

Based on the adaptive learning design (14), the virtual control $s_{i,1}$ is considered as

$$s_{i,1} = \frac{1}{o_i} \left(-\Lambda_{i,1} \delta_{i,1} - \frac{\hat{\Theta}_{i,1}}{2\eta_{i,1}} v_{i,1} \phi_{i,1}^T \phi_{i,1} - o_i \lambda_{i,2} + \sum_{j \in \mathcal{N}_i} a_{i,j} x_{j,2} + b_i \dot{x}_0 \right). \quad (15)$$

Substituting (15) into (12) yields

$$\begin{aligned} \dot{\delta}_{i,1} = & - \sum_{j \in \mathcal{N}_i} a_{i,j} \left[f_{j,1}^{\sigma_j(t)}(x_{j,1}) + g_{j,1}^{\sigma_j(t)}(x_{j,1}(t - \tau_{j,1}^{\sigma_j(t)})) \right] \\ & + o_i \left[f_{i,1}^{\sigma_i(t)}(x_{i,1}) + g_{i,1}^{\sigma_i(t)}(x_{i,1}(t - \tau_{i,1}^{\sigma_i(t)})) \right] \\ & - \Lambda_{i,1} \delta_{i,1} + o_i \delta_{i,2} + o_i \tilde{\alpha}_{i,1} - \frac{\hat{\Theta}_{i,1}}{2\eta_{i,1}} v_{i,1} \phi_{i,1}^T \phi_{i,1}. \end{aligned} \quad (16)$$

Therefore, combining (6) and (16) yields

$$\begin{aligned} \dot{v}_{i,1} = & - \sum_{j \in \mathcal{N}_i} a_{i,j} \left[f_{j,1}^{\sigma_j(t)}(x_{j,1}) + g_{j,1}^{\sigma_j(t)}(x_{j,1}(t - \tau_{j,1}^{\sigma_j(t)})) \right] + o_i \left[f_{i,1}^{\sigma_i(t)}(x_{i,1}) + g_{i,1}^{\sigma_i(t)}(x_{i,1}(t - \tau_{i,1}^{\sigma_i(t)})) \right] \\ & - \Lambda_{i,1} v_{i,1} + o_i v_{i,2} - \frac{\hat{\Theta}_{i,1}}{2\eta_{i,1}} v_{i,1} \phi_{i,1}^T \phi_{i,1}. \end{aligned} \quad (17)$$

Unlike constant input delays or communication delays, the time-varying full-state delays in (1) enter each strict-feedback layer of every agent and differ across agents and modes. As a result, the local bipartite error dynamics (12) involve both the current and delayed neighbor states, and cannot be transformed into a delay-free form by simple state augmentation. On the analysis side, the Lyapunov-Krasovskii functional must contain mode-dependent integral terms whose derivatives generate delay-derivative-dependent boundary terms. The boundedness condition in Assumption 3 is therefore essential to dominate these terms and retain a negative definite contribution in the Lyapunov derivative.

Consider the Lyapunov-Krasovskii candidate for the first subsystem of the i -th agent with switching mode $\sigma_i(t)$ and all its neighbors, and the j -th agent with switching mode $\sigma_j(t)$ for $j \in \mathcal{N}_i$, as

$$V_{i,1}^{\sigma_i(t)} = \hat{V}_{i,1}^{\sigma_i(t)} + \check{V}_{i,1}^{\sigma_i(t)} + \sum_{j \in \mathcal{N}_i} \check{V}_{j,1}^{\sigma_j(t)}, \quad (18)$$

with $\hat{V}_{i,1}^{\sigma_i(t)} = \frac{1}{2} v_{i,1}^2 + \frac{1}{2} \hat{\Theta}_{i,1}^2$ and

$$\check{V}_{i,1}^{\sigma_i(t)} = \int_{t-\tau_{i,1}^{\sigma_i(t)}}^t e^{\varpi_i [s-t+\tau_{i,1}^{\sigma_i(t)}]} \left[g_{i,1}^{\sigma_i(s)}(x_{i,1}(s)) \right]^2 ds,$$

where $\varpi_i > 0$. With (14) and (17), $\dot{V}_{i,1}^{\sigma_i(t)}$ satisfies

$$\begin{aligned} \dot{V}_{i,1}^{\sigma_i(t)} = & v_{i,1} \left[o_i f_{i,1}^{\sigma_i(t)}(x_{i,1}) + o_i g_{i,1}^{\sigma_i(t)}(x_{i,1}(t - \tau_{i,1}^{\sigma_i(t)})) + o_i v_{i,2} - \Lambda_{i,1} v_{i,1} - \sum_{j \in N_i} a_{i,j} f_{j,1}^{\sigma_j(t)}(x_{j,1}) \right. \\ & \left. - \sum_{j \in N_i} a_{i,j} g_{j,1}^{\sigma_j(t)}(x_{j,1}(t - \tau_{j,1}^{\sigma_j(t)})) \right] - \frac{1}{2\eta_{i,1}^2} \Theta_{i,1} v_{i,1}^2 \phi_{i,1}^T \phi_{i,1} + \Gamma_{i,1} \tilde{\Theta}_{i,1} \hat{\Theta}_{i,1}. \end{aligned} \quad (19)$$

One can observe that the term $\dot{V}_{i,1}^{\sigma_i(t)} = \frac{1}{2} v_{i,1}^2 + \frac{1}{2} \tilde{\Theta}_{i,1}^2$ with a switching index $\sigma_i(t)$ does not explicitly contain switching signals in its expression. However, the overall switching behavior in system (1) leads to the switching signals such as $f_{i,1}^{\sigma_i(t)}$, $g_{i,1}^{\sigma_i(t)}$, and $\tau_{i,1}^{\sigma_i(t)}$ involved by the dynamics (19) of $\dot{V}_{i,1}^{\sigma_i(t)}$. So the term $\dot{V}_{i,1}^{\sigma_i(t)}$ is also indexed by $\sigma_i(t)$.

Differentiating $\check{V}_{i,1}^{\sigma_i(t)} + \sum_{j \in N_i} \check{V}_{j,1}^{\sigma_j(t)}$ yields

$$\begin{aligned} & \dot{\check{V}}_{i,1}^{\sigma_i(t)} + \sum_{j \in N_i} \dot{\check{V}}_{j,1}^{\sigma_j(t)} \\ = & e^{\varpi_i \tau_{i,1}^{\sigma_i(t)}} \left[g_{i,1}^{\sigma_i(t)}(x_{i,1}) \right]^2 - \varpi_i (1 - \tau_{i,1}^{\sigma_i(t)}) \check{V}_{i,1}^{\sigma_i(t)} - (1 - \tau_{i,1}^{\sigma_i(t)}) \left[g_{i,1}^{\sigma_i(t)}(x_{i,1}(t - \tau_{i,1}^{\sigma_i(t)})) \right]^2 \\ & + \sum_{j \in N_i} \left\{ e^{\varpi_j \tau_{j,1}^{\sigma_j(t)}} \left[g_{j,1}^{\sigma_j(t)}(x_{j,1}) \right]^2 - \varpi_j (1 - \tau_{j,1}^{\sigma_j(t)}) \check{V}_{j,1}^{\sigma_j(t)} - (1 - \tau_{j,1}^{\sigma_j(t)}) \left[g_{j,1}^{\sigma_j(t)}(x_{j,1}(t - \tau_{j,1}^{\sigma_j(t)})) \right]^2 \right\}, \end{aligned} \quad (20)$$

Combining the results in (20) and (19) yields

$$\begin{aligned} \dot{V}_{i,1}^{\sigma_i(t)} = & v_{i,1} \left\{ o_i f_{i,1}^{\sigma_i(t)}(x_{i,1}) + o_i g_{i,1}^{\sigma_i(t)}(x_{i,1}(t - \tau_{i,1}^{\sigma_i(t)})) - \sum_{j \in N_i} a_{i,j} [f_{j,1}^{\sigma_j(t)}(x_{j,1}) + g_{j,1}^{\sigma_j(t)}(x_{j,1}(t - \tau_{j,1}^{\sigma_j(t)}))] \right. \\ & \left. + o_i v_{i,2} - \Lambda_{i,1} v_{i,1} \right\} - \frac{1}{2\eta_{i,1}^2} \Theta_{i,1} v_{i,1}^2 \phi_{i,1}^T \phi_{i,1} + \Gamma_{i,1} \tilde{\Theta}_{i,1} \hat{\Theta}_{i,1} + e^{\varpi_i \tau_{i,1}^{\sigma_i(t)}} [g_{i,1}^{\sigma_i(t)}(x_{i,1})]^2 - \varpi_i (1 - \tau_{i,1}^{\sigma_i(t)}) \check{V}_{i,1}^{\sigma_i(t)} \\ & - (1 - \tau_{i,1}^{\sigma_i(t)}) [g_{i,1}^{\sigma_i(t)}(x_{i,1}(t - \tau_{i,1}^{\sigma_i(t)}))]^2 + \sum_{j \in N_i} \left\{ e^{\varpi_j \tau_{j,1}^{\sigma_j(t)}} [g_{j,1}^{\sigma_j(t)}(x_{j,1})]^2 - \varpi_j (1 - \tau_{j,1}^{\sigma_j(t)}) \check{V}_{j,1}^{\sigma_j(t)} \right. \\ & \left. - (1 - \tau_{j,1}^{\sigma_j(t)}) [g_{j,1}^{\sigma_j(t)}(x_{j,1}(t - \tau_{j,1}^{\sigma_j(t)}))]^2 \right\}. \end{aligned} \quad (21)$$

Applying Assumption 3 and Young's inequality yields $e^{\varpi_i \tau_{i,1}^{\sigma_i(t)}} [g_{i,1}^{\sigma_i(t)}(x_{i,1})]^2 \geq e^{\varpi_i \tau_{i,1}^{\sigma_i(t)}} [g_{i,1}^{\sigma_i(t)}(x_{i,1})]^2$ and

$$\begin{aligned} \frac{o_i^2 v_{i,1}^2}{4(1 - \bar{\tau}_{d,i})} & \geq o_i v_{i,1} g_{i,1}^{\sigma_i(t)}(x_{i,1}(t - \tau_{i,1}^{\sigma_i(t)})) - (1 - \bar{\tau}_{d,i}) [g_{i,1}^{\sigma_i(t)}(x_{i,1}(t - \tau_{i,1}^{\sigma_i(t)}))]^2, \\ \frac{v_{i,1}^2}{4(1 - \bar{\tau}_{d,j})} & \geq -v_{i,1} a_{i,j} g_{j,1}^{\sigma_j(t)}(x_{j,1}(t - \tau_{j,1}^{\sigma_j(t)})) - (1 - \bar{\tau}_{d,j}) [g_{j,1}^{\sigma_j(t)}(x_{j,1}(t - \tau_{j,1}^{\sigma_j(t)}))]^2. \end{aligned}$$

Therefore, we can obtain

$$\begin{aligned} \dot{V}_{i,1}^{\sigma_i(t)} \leq & -\Lambda_{i,1} v_{i,1}^2 + o_i v_{i,1} v_{i,2} + v_{i,1} H_{i,1}^{\sigma_i(t)} + \left[1 - 2 \tanh^2 \left(\frac{v_{i,1}}{\chi_{i,1}} \right) \right] G_{i,1}^{\sigma_i(t)} \\ & - \varpi_i (1 - \bar{\tau}_{d,i}) \check{V}_{i,1}^{\sigma_i(t)} - \sum_{j \in N_i} \varpi_j (1 - \bar{\tau}_{d,j}) \check{V}_{j,1}^{\sigma_j(t)} - \frac{1}{2\eta_{i,1}^2} \Theta_{i,1} v_{i,1}^2 \phi_{i,1}^T \phi_{i,1} + \Gamma_{i,1} \tilde{\Theta}_{i,1} \hat{\Theta}_{i,1}. \end{aligned} \quad (22)$$

With the Young's inequality, we can obtain

$$\begin{aligned} v_{i,1} H_{i,1}^{\sigma_i(t)}(X_{i,1}) & = v_{i,1} \left[\left(\theta_{i,1}^{\sigma_i(t)} \right)^T \phi_{i,1}(X_{i,1}) + \varepsilon_{i,1}^{\sigma_i(t)}(X_{i,1}) \right] \leq \frac{\Theta_{i,1} v_{i,1}^2 \phi_{i,1}^T \phi_{i,1}}{2\eta_{i,1}} + \frac{v_{i,1}^2 + \bar{\varepsilon}_{i,1}^2 + \eta_{i,1}}{2}, \\ \tilde{\Theta}_{i,1} \hat{\Theta}_{i,1} & \leq \frac{1}{2} \Theta_{i,1}^2 - \frac{1}{2} \tilde{\Theta}_{i,1}^2, \end{aligned}$$

where $\bar{\varepsilon}_{i,1} = \max_{q_{i,k} \in Q_i} \bar{\varepsilon}_{i,1}^{q_{i,k}}$. Therefore, $\dot{V}_{i,1}^{\sigma_i(t)}$ satisfies

$$\begin{aligned} \dot{V}_{i,1}^{\sigma_i(t)} \leq & -\left(\Lambda_{i,1} - \frac{1}{2} \right) v_{i,1}^2 + \bar{\Lambda}_{i,2} v_{i,1} v_{i,2} + \left[1 - 2 \tanh^2 \left(\frac{v_{i,1}}{\chi_{i,1}} \right) \right] G_{i,1}^{\sigma_i(t)} \\ & - \varpi_i (1 - \bar{\tau}_{d,i}) \check{V}_{i,1}^{\sigma_i(t)} - \sum_{j \in N_i} \varpi_j (1 - \bar{\tau}_{d,j}) \check{V}_{j,1}^{\sigma_j(t)} + \frac{\Gamma_{i,1}}{2} \Theta_{i,1}^2 - \frac{\Gamma_{i,1}}{2} \tilde{\Theta}_{i,1}^2 + \frac{\eta_{i,1} + \varepsilon_{i,1}^2}{2}. \end{aligned} \quad (23)$$

The time-varying delay $\tau_{i,l}^{\sigma_i(t)}(t)$ thus affects both the protocol design and the analysis: it appears explicitly in the dynamics of $\delta_{i,l}$ in (12), and the derivative $\dot{\tau}_{i,l}^{\sigma_i(t)}(t)$ enters the Lyapunov-Krasovskii derivative through the integral term $\check{V}_{i,1}^{\sigma_i(t)} + \sum_{j \in \mathcal{N}_i} \check{V}_{j,1}^{\sigma_j(t)}$ in (18).

Assumption 3 guarantees that these delay terms can be compensated by $-\varpi_i(1 - \bar{\tau}_{d,i})\check{V}_{i,1}^{\sigma_i(t)} - \sum_{j \in \mathcal{N}_i} \varpi_j(1 - \bar{\tau}_{d,j})\check{V}_{j,1}^{\sigma_j(t)}$ as in (23).

Inequality (23) implies that the derivative of $V_{i,1}^{\sigma_i(t)}$ is negative definite with respect to $v_{i,1}$ outside a bounded neighborhood, provided that the design gains $\Lambda_{i,1}$, $\Gamma_{i,1}$, and $\eta_{i,1}$ are chosen sufficiently large. Together with the boundedness of the RNFNN approximation error and the time-delay related terms, this establishes uniform ultimate boundedness (UUB) of the first-layer error $v_{i,1}$, which will be included later in the detailed discussion for stability analysis.

It is worth emphasizing how the Lyapunov-Krasovskii functional is constructed to cope with multiple distinct state delays across the agents. For each follower i , the candidate functional consists of two parts: a standard Lyapunov term that depends on the compensated error $v_{i,l}$ and the adaptive parameter errors, and an integral term that explicitly involves the delayed state $x_i(t - \tau_{i,l}^{\sigma_i(t)})$ through the mode-dependent function $g_{i,l}^{\sigma_i(\cdot)}$; see (18)–(22). The integral term is weighted by an exponentially decaying kernel and is defined over the interval $[t - \tau_{i,l}^{\sigma_i(t)}, t]$, so that the contribution of the delay can be upper-bounded using **Assumption 3** on the delay magnitude and its derivative. In addition, for each neighbour $j \in \mathcal{N}_i$, a similar integral term is included to capture the effect of the neighbour's delayed state dynamics under its own switching signal $\sigma_j(t)$. In this way, the overall functional aggregates the delayed contributions of all agents and modes, which enables us to derive mode-dependent dissipation inequalities and to combine them with the ADT condition so as to guarantee global stability despite asynchronous switching and heterogeneous, time-varying state delays.

Remark 7. The filter parameters play a key role in shaping the bipartite synchronization performance. The time constants $\omega_{i,l} > 0$ in (5) determine the bandwidth of the DSC filters: larger values reduce the lag between $s_{i,l}$ and $\alpha_{i,l}$ and hence the filtered errors $\tilde{\alpha}_{i,l}$, while very large values may amplify high-frequency components. The gains $\Lambda_{i,l}$ and $\bar{\Lambda}_{i,l}$ in (6)–(8) govern the convergence rate of the compensation signals $\beta_{i,l}$ and lead to the negative definite terms $-\Lambda_{i,l}v_{i,1}^2$ in the Lyapunov inequalities in (23). Finally, the parameters $\rho_{i,l}$ and $\bar{\rho}_{i,n_i}$ in (10) determine the responsiveness of the second-layer filter to the dead-zone mapping. The condition $\bar{\rho}_{i,n_i} \geq 1 + \max\{c_{i,4}, c_{i,6}\}$ ensures that the dead-zone-induced terms can be dominated by the damping in the $\lambda_{i,l}$ dynamics, which is critical for the boundedness of the bipartite synchronization error.

3.2.2. Step $l(2 \leq l \leq n_i - 1)$

Based on (11), differentiating $\delta_{i,l}$ yields

$$\dot{\delta}_{i,l} = \dot{x}_{i,l} - \dot{\alpha}_{i,l-1} - \dot{\lambda}_{i,l}. \quad (24)$$

Substituting (1) and (10) into (24) yields

$$\dot{\delta}_{i,l} = f_{i,l}^{\sigma_i(t)}(\tilde{x}_{i,l}) + g_{i,l}^{\sigma_i(t)}(\tilde{x}_{i,l}(t - \tau_{i,l}^{\sigma_i(t)})) + \delta_{i,l+1} + \tilde{\alpha}_{i,l} + s_{i,l} - \dot{\alpha}_{i,l-1} + \rho_{i,l}\lambda_{i,l}. \quad (25)$$

Similar to Step 1, consider

$$\begin{aligned} H_{i,l}^{\sigma_i(t)}(X_{i,l}) &= f_{i,l}^{\sigma_i(t)}(\tilde{x}_{i,l}) + \frac{1}{4(1 - \bar{\tau}_{d,i})}v_{i,l} + 2\tanh^2\left(\frac{v_{i,l}}{\chi_{i,l}}\right)\frac{G_{i,l}^{\sigma_i(t)}(\tilde{x}_{i,l})}{v_{i,l}}, \\ G_{i,l}^{\sigma_i(t)}(\tilde{x}_{i,l}) &= e^{\varpi_i \bar{\tau}_i} \left[g_{i,l}^{\sigma_i(t)}(\tilde{x}_{i,l}) \right]^2 \end{aligned} \quad (26)$$

with $X_{i,l} = [x_{i,1}, \dots, x_{i,l}, v_{i,l}]^T$ and $\chi_{i,l} > 0$. Applying RBFNN to approximate $H_{i,l}^{\sigma_i(t)}(X_{i,l})$ yields

$$H_{i,l}^{q_{i,k}}(\cdot) = \left(\theta_{i,l}^{q_{i,k}} \right)^T \phi_{i,l}(\cdot) + \varepsilon_{i,l}^{q_{i,k}}(\cdot), \forall q_{i,k} \in \mathcal{Q}_i \quad (27)$$

on compact set $\Omega_{i,l}^X$ with the ideal weight defined as $\theta_{i,l}^{q_{i,k}} = \arg \min_{\theta \in \Omega_{i,l}^\theta} \sup_{X \in \Omega_{i,l}^X} \|H_{i,l}^{q_{i,k}}(X) - \theta^T \phi_{i,l}(X)\|$, bounded residual $|\varepsilon_{i,l,k_i}^*(X_{i,l})| \leq \varepsilon_{i,l}$

with accuracy level $\varepsilon_{i,l} > 0$, and the $N_{i,l}$ -dimensional radial basis function $\phi_{i,l}(X_{i,l}) = [\phi_{i,l,1}(X_{i,l}), \dots, \phi_{i,l,N_{i,l}}(X_{i,l})]^T$. For the unknown ideal weight vector $\theta_{i,l}^{q_{i,k}}$, consider the following adaptive law design to compensate for the term $\Theta_{i,l} = \max_{q_{i,k} \in \mathcal{Q}_i} \|\theta_{i,l}^{q_{i,k}}\|^2$ as

$$\dot{\hat{\Theta}}_{i,l} = -\Gamma_{i,l}\hat{\Theta}_{i,l} + \frac{1}{2\eta_{i,l}^2}v_{i,l}^2\phi_{i,l}^T\phi_{i,l}, \quad \hat{\Theta}_{i,l}(0) \geq 0, \quad (28)$$

with design parameters $\Gamma_{i,l}$, $\eta_{i,l}$ and learning error $\tilde{\Theta}_{i,l} = \Theta_{i,l} - \hat{\Theta}_{i,l}$.

Then, virtual control is designed as

$$s_{i,l} = -\Lambda_{i,l-1}\delta_{i,l-1} - \Lambda_{i,l}\delta_{i,l} - \rho_{i,l}\lambda_{i,l} - \frac{\hat{\Theta}_{i,l}}{2\eta_{i,l}}v_{i,l}\phi_{i,l}^T\phi_{i,l} + \dot{\alpha}_{i,l-1}. \quad (29)$$

Inserting (29) into (30), we have

$$\dot{\delta}_{i,l} = f_{i,l}^{\sigma_i(t)}(\bar{x}_{i,l}) + g_{i,l}^{\sigma_i(t)}\left(\bar{x}_{i,l}\left(t - \tau_{i,l}^{\sigma_i(t)}\right)\right) + \tilde{\alpha}_{i,l} - \underline{\Lambda}_{i,l-1}\delta_{i,l-1} - \Lambda_{i,l}\delta_{i,l} + \delta_{i,l+1} - \frac{\hat{\Theta}_{i,l}}{2\eta_{i,l}}v_{i,l}\phi_{i,l}^T\phi_{i,l}, \quad (30)$$

From (7) and (30), we can obtain

$$\dot{v}_{i,l} = f_{i,l}^{\sigma_i(t)}(\bar{x}_{i,l}) + g_{i,l}^{\sigma_i(t)}\left(\bar{x}_{i,l}\left(t - \tau_{i,l}^{\sigma_i(t)}\right)\right) - \underline{\Lambda}_{i,l-1}v_{i,l-1} - \Lambda_{i,l}v_{i,l} + v_{i,l+1} - \frac{\hat{\Theta}_{i,l}}{2\eta_{i,l}}v_{i,l}\phi_{i,l}^T\phi_{i,l}. \quad (31)$$

Denote the Lyapunov-Krasovskii candidate for the $l - th$ subsystem of $i - th$ agent with switching mode $\sigma_i(t)$ as

$$V_{i,l}^{\sigma_i(t)} = \hat{V}_{i,l}^{\sigma_i(t)} + \check{V}_{i,l}^{\sigma_i(t)}, \quad (32)$$

with $\hat{V}_{i,l}^{\sigma_i(t)} = \frac{1}{2}v_{i,l}^2 + \frac{1}{2}\tilde{\Theta}_{i,l}^2$ and $\check{V}_{i,l}^{\sigma_i(t)} = \int_{t-\tau_{i,l}^{\sigma_i(t)}}^t e^{\varpi_i[s-t+\tau_{i,l}^{\sigma_i(t)}]} \left[g_{i,l}^{\sigma_i(t)}(\bar{x}_{i,l}(s)) \right]^2 ds$. With the facts in (28) and (31), by differentiating $\hat{V}_{i,l}^{\sigma_i(t)}$, we have

$$\begin{aligned} \dot{\hat{V}}_{i,l}^{\sigma_i(t)} &= v_{i,l}\dot{v}_{i,l} - \tilde{\Theta}_{i,l}\dot{\tilde{\Theta}}_{i,l} = v_{i,l}\left(-\underline{\Lambda}_{i,l-1}v_{i,l-1} - \Lambda_{i,l}v_{i,l} + v_{i,l+1}\right) + v_{i,l}\left[f_{i,l}^{\sigma_i(t)}(\bar{x}_{i,l}) + g_{i,l}^{\sigma_i(t)}\left(\bar{x}_{i,l}\left(t - \tau_{i,l}^{\sigma_i(t)}\right)\right)\right] \\ &\quad - \frac{\Theta_{i,l}}{2\eta_{i,l}^2}v_{i,l}^2\phi_{i,l}^T\phi_{i,l} + \Gamma_{i,l}\tilde{\Theta}_{i,l}\hat{\Theta}_{i,l}. \end{aligned} \quad (33)$$

Taking the derivative of $\check{V}_{i,l}^{\sigma_i(t)}$, we can obtain

$$\dot{\check{V}}_{i,l}^{\sigma_i(t)} = e^{\varpi_i\tau_{i,l}^{\sigma_i(t)}} \left[g_{i,l}^{\sigma_i(t)}(\bar{x}_{i,l}) \right]^2 - \varpi_i \left(1 - \hat{\tau}_{i,l}^{\sigma_i(t)}\right) \check{V}_{i,l}^{\sigma_i(t)} - \left(1 - \hat{\tau}_{i,l}^{\sigma_i(t)}\right) \left[g_{i,l}^{\sigma_i(t)}\left(\bar{x}_{i,l}\left(t - \tau_{i,l}^{\sigma_i(t)}\right)\right) \right]^2. \quad (34)$$

Collecting results in (34) and (33) yields

$$\begin{aligned} \dot{V}_{i,l}^{\sigma_i(t)} &= v_{i,l}\left(-\underline{\Lambda}_{i,l-1}v_{i,l-1} - \Lambda_{i,l}v_{i,l} + v_{i,l+1}\right) + v_{i,l}\left[f_{i,l}^{\sigma_i(t)}(\bar{x}_{i,l}) + g_{i,l}^{\sigma_i(t)}\left(\bar{x}_{i,l}\left(t - \tau_{i,l}^{\sigma_i(t)}\right)\right)\right] \\ &\quad - \frac{\Theta_{i,l}}{2\eta_{i,l}^2}v_{i,l}^2\phi_{i,l}^T\phi_{i,l} + \Gamma_{i,l}\tilde{\Theta}_{i,l}\hat{\Theta}_{i,l} + e^{\varpi_i\tau_{i,l}^{\sigma_i(t)}} \left[g_{i,l}^{\sigma_i(t)}(\bar{x}_{i,l}) \right]^2 - \varpi_i \left(1 - \hat{\tau}_{i,l}^{\sigma_i(t)}\right) \check{V}_{i,l}^{\sigma_i(t)} \\ &\quad - \left(1 - \hat{\tau}_{i,l}^{\sigma_i(t)}\right) \left[g_{i,l}^{\sigma_i(t)}\left(\bar{x}_{i,l}\left(t - \tau_{i,l}^{\sigma_i(t)}\right)\right) \right]^2. \end{aligned} \quad (35)$$

Based on Assumption 3 and Young's inequality, we have

$$\begin{aligned} e^{\varpi_i\bar{\tau}_i} &\geq e^{\varpi_i\tau_{i,l}^{\sigma_i(t)}}, \\ \frac{1}{4(1-\bar{\tau}_{d,i})}v_{i,l}^2 &\geq v_{i,l}g_{i,l}^{\sigma_i(t)}\left(\bar{x}_{i,l}\left(t - \tau_{i,l}^{\sigma_i(t)}\right)\right) - \left(1 - \hat{\tau}_{i,l}^{\sigma_i(t)}\right) \left[g_{i,l}^{\sigma_i(t)}\left(\bar{x}_{i,l}\left(t - \tau_{i,l}^{\sigma_i(t)}\right)\right) \right]^2. \end{aligned}$$

Therefore, we can obtain

$$\begin{aligned} \dot{V}_{i,l}^{\sigma_i(t)} &\leq -\underline{\Lambda}_{i,l-1}v_{i,l-1}v_{i,l} - \Lambda_{i,l}v_{i,l}^2 + v_{i,l}v_{i,l+1} - \frac{v_{i,l}^2}{2\eta_{i,l}}\Theta_{i,l}\phi_{i,l}^T\phi_{i,l} + \Gamma_{i,l}\tilde{\Theta}_{i,l}\hat{\Theta}_{i,l} + v_{i,l}H_{i,l}^{\sigma_i(t)}(X_{i,l}) \\ &\quad + \left[1 - 2\tanh^2\left(\frac{v_{i,l}}{\chi_{i,l}}\right)\right]G_{i,l}^{\sigma_i(t)} - \varpi_i(1-\bar{\tau}_{d,i})\check{V}_{i,l}^{\sigma_i(t)}. \end{aligned} \quad (36)$$

Employing Young's inequality, we can obtain

$$v_{i,l}H_{i,l}^{\sigma_i(t)}(X_{i,l}) \leq \frac{v_{i,l}^2}{2\eta_{i,l}}\Theta_{i,l}\phi_{i,l}^T\phi_{i,l} + \frac{\eta_{i,l} + v_{i,l}^2 + \epsilon_{i,l}^2}{2}, \quad \tilde{\Theta}_{i,l}\hat{\Theta}_{i,l} \leq \frac{1}{2}\Theta_{i,l}^2 - \frac{1}{2}\tilde{\Theta}_{i,l}^2. \quad (37)$$

Substituting (28) and (37) into (35), $\dot{V}_{i,l,k_i}$ further satisfies

$$\begin{aligned} \dot{V}_{i,l}^{\sigma_i(t)} &\leq -\left(\Lambda_{i,l} - \frac{1}{2}\right)v_{i,l}^2 - \underline{\Lambda}_{i,l-1}v_{i,l-1}v_{i,l} + v_{i,l}v_{i,l+1} - \frac{\Gamma_{i,l}}{2}\tilde{\Theta}_{i,l}^2 + \frac{\Gamma_{i,l}\Theta_{i,l}^2 + \eta_{i,l} + \epsilon_{i,l}^2}{2} - \varpi_i(1-\bar{\tau}_{d,i})\check{V}_{i,l}^{\sigma_i(t)} \\ &\quad + \left[1 - 2\tanh^2\left(\frac{v_{i,l}}{\chi_{i,l}}\right)\right]G_{i,l}^{\sigma_i(t)}. \end{aligned} \quad (38)$$

3.2.3. Step n_i

From (11), differentiating δ_{i,n_i} yields

$$\dot{\delta}_{i,n_i} = \dot{x}_{i,n_i} - \dot{\alpha}_{i,n_i-1} - \dot{\lambda}_{i,n_i}. \quad (39)$$

Substituting (1) and (10) into (39), we have

$$\dot{\delta}_{i,n_i} = f_{i,n_i}^{\sigma_i(t)}(x_i) + g_{i,n_i}^{\sigma_i(t)}\left(x_i\left(t - \tau_{i,n_i}^{\sigma_i(t)}\right)\right) + u_i - \dot{\alpha}_{i,n_i-1} + \bar{\rho}_{i,n_i}\rho_{i,n_i}\lambda_{i,n_i}. \quad (40)$$

Denote $X_{i,n_i} = [x_{i,1}, \dots, x_{i,n_i}, v_{i,n_i}, \lambda_{i,n_i}]^T$ and

$$\begin{aligned} H_{i,n_i}^{\sigma_i(t)}(X_i) &= f_{i,n_i}^{\sigma_i(t)}(x_i) + \frac{1}{4(1 - \tau_{i,n_i})} v_{i,n_i} + (\bar{\rho}_{i,n_i} - 1) \rho_{i,n_i} \lambda_{i,n_i} + 2 \tanh^2\left(\frac{v_{i,n_i}}{X_{i,n_i}}\right) \frac{G_{i,n_i}^{\sigma_i(t)}(x_i)}{v_{i,n_i}}, \\ G_{i,n_i}^{\sigma_i(t)}(x_i) &= e^{\varpi_i \bar{\tau}_i} \left[g_{i,n_i}^{\sigma_i(t)}(x_i) \right]^2 \end{aligned} \quad (41)$$

with $X_{i,n_i} > 0$. To address the unknown functions in $H_{i,n_i}^{\sigma_i(t)}(\cdot)$, utilizing RBFNN yields

$$H_{i,n_i}^{q_{i,k}}(\cdot) = \left(\theta_{i,n_i}^{q_{i,k}} \right)^T \phi_{i,n_i}(\cdot) + \varepsilon_{i,n_i}^{q_{i,k}}(\cdot), \forall q_{i,k} \in \mathcal{Q}_i,$$

which holds on the compact set Ω_{i,n_i}^X with the approximation residual satisfying $\left| \varepsilon_{i,n_i}^{q_{i,k}}(X_{i,n_i}) \right| \leq \varepsilon_{i,n_i}$, the accuracy level $\varepsilon_{i,n_i} > 0$, and the ideal weight

$$\theta_{i,n_i}^{q_{i,k}} = \arg \min_{\theta \in \Omega_{i,n_i}^\theta} \sup_{X \in \Omega_{i,n_i}^X} \left\| H_{i,n_i}^{q_{i,k}}(X) - \theta^T \phi_{i,n_i}(X) \right\|, \quad (42)$$

and the N_{i,n_i} -dimensional radial basis function

$$\phi_{i,n_i}(X_{i,n_i}) = [\phi_{i,n_i,1}(X_{i,n_i}), \dots, \phi_{i,n_i,N_{i,n_i}}(X_{i,n_i})]^T. \quad (43)$$

It should be noted that $\theta_{i,n_i}^{q_{i,k}}$ is unknown. Denote $\Theta_{i,n_i} = \max_{q_{i,k} \in \mathcal{Q}_i} \left\| \theta_{i,n_i}^{q_{i,k}} \right\|^2$ with its adaptive estimation $\hat{\Theta}_{i,n_i}$, where the adaptive law is constructed as

$$\dot{\hat{\Theta}}_{i,n_i} = -\Gamma_{i,n_i} \hat{\Theta}_{i,n_i} + \frac{1}{2\eta_{i,n_i}^2} v_{i,n_i}^2 \phi_{i,n_i}^T \phi_{i,n_i}, \quad (44)$$

with initial condition $\hat{\Theta}_{i,n_i}(0) \geq 0$, positive learning parameters Γ_{i,n_i} and η_{i,n_i} , and learning error $\tilde{\Theta}_{i,n_i} = \Theta_{i,n_i} - \hat{\Theta}_{i,n_i}$. The controller is developed as

$$u_i = -\underline{\Lambda}_{i,n_i-1} \delta_{i,n_i-1} - \Lambda_{i,n_i} \delta_{i,n_i} - \rho_{i,n_i} \lambda_{i,n_i} - \frac{\hat{\Theta}_{i,n_i}}{2\eta_{i,n_i}} v_{i,n_i} \phi_{i,n_i}^T \phi_{i,n_i} + \dot{\alpha}_{i,n_i-1}, \quad (45)$$

with $\Lambda_{i,n_i} > 0$. Inserting (45) into (40), we have

$$\dot{\delta}_{i,n_i} = f_{i,n_i}^{\sigma_i(t)}(x_i) + g_{i,n_i}^{\sigma_i(t)}\left(x_i\left(t - \tau_{i,n_i}^{\sigma_i(t)}\right)\right) + (\bar{\rho}_{i,n_i} - 1) \rho_{i,n_i} \lambda_{i,n_i} - \frac{\hat{\Theta}_{i,n_i}}{2\eta_{i,n_i}} v_{i,n_i} \phi_{i,n_i}^T \phi_{i,n_i} - \underline{\Lambda}_{i,n_i-1} \delta_{i,n_i-1} - \Lambda_{i,n_i} \delta_{i,n_i}. \quad (46)$$

From (46) and (8), we can obtain

$$\dot{v}_{i,n_i} = f_{i,n_i}^{\sigma_i(t)}(x_i) + g_{i,n_i}^{\sigma_i(t)}\left(x_i\left(t - \tau_{i,n_i}^{\sigma_i(t)}\right)\right) + (\bar{\rho}_{i,n_i} - 1) \rho_{i,n_i} \lambda_{i,n_i} - \frac{\hat{\Theta}_{i,n_i}}{2\eta_{i,n_i}} v_{i,n_i} \phi_{i,n_i}^T \phi_{i,n_i} - \underline{\Lambda}_{i,n_i-1} v_{i,n_i-1} - \Lambda_{i,n_i} v_{i,n_i}. \quad (47)$$

For the $n_i - th$ subsystem of $i - th$ agent with switching mode $\sigma_i(t)$, consider the Lyapunov-Krasovskii functional

$$V_{i,n_i}^{\sigma_i(t)} = \hat{V}_{i,n_i}^{\sigma_i(t)} + \check{V}_{i,n_i}^{\sigma_i(t)}, \quad (48)$$

with

$$\check{V}_{i,n_i}^{\sigma_i(t)} = \int_{t-\tau_{i,n_i}^{\sigma_i(t)}}^t e^{\varpi_i[s-t+\tau_{i,n_i}^{\sigma_i(t)}]} \left[g_{i,n_i}^{\sigma_i(t)}(x_i(s)) \right]^2 ds, \quad \hat{V}_{i,n_i}^{\sigma_i(t)} = \frac{1}{2} v_{i,n_i}^2 + \frac{1}{2} \hat{\Theta}_{i,n_i}^2. \quad (49)$$

With (44) and (47), differentiating $\hat{V}_{i,n_i}^{\sigma_i(t)}$ yields

$$\begin{aligned} \dot{\hat{V}}_{i,n_i}^{\sigma_i(t)} &= v_{i,n_i} \dot{v}_{i,n_i} - \tilde{\Theta}_{i,n_i} \dot{\hat{\Theta}}_{i,n_i}, \\ &= v_{i,n_i} \left[f_{i,n_i}^{\sigma_i(t)}(x_i) + g_{i,n_i}^{\sigma_i(t)}\left(x_i\left(t - \tau_{i,n_i}^{\sigma_i(t)}\right)\right) \right] + (\bar{\rho}_{i,n_i} - 1) \rho_{i,n_i} v_{i,n_i} \lambda_{i,n_i} \\ &\quad - \frac{\Theta_{i,n_i}}{2\eta_{i,n_i}^2} v_{i,n_i}^2 \phi_{i,n_i}^T \phi_{i,n_i} + \Gamma_{i,n_i} \tilde{\Theta}_{i,n_i} \hat{\Theta}_{i,n_i} - \Lambda_{i,n_i} v_{i,n_i}^2 - \underline{\Lambda}_{i,n_i-1} v_{i,n_i-1} v_{i,n_i}. \end{aligned} \quad (50)$$

The derivative of $\check{V}_{i,n_i}^{\sigma_i(t)}$ can be expressed as

$$\dot{\check{V}}_{i,n_i}^{\sigma_i(t)} = e^{\varpi_i \tau_{i,n_i}^{\sigma_i(t)}} \left[g_{i,n_i}^{\sigma_i(t)}(x_i(t)) \right]^2 - \varpi_i \left(1 - \tau_{i,n_i}^{\sigma_i(t)} \right) \check{V}_{i,n_i}^{\sigma_i(t)} - \left(1 - \tau_{i,n_i}^{\sigma_i(t)} \right) \left[g_{i,n_i}^{\sigma_i(t)}\left(x_i\left(t - \tau_{i,n_i}^{\sigma_i(t)}\right)\right) \right]^2. \quad (51)$$

Combining the results in (51) and (50), we can obtain

$$\dot{V}_{i,n_i}^{\sigma_i(t)} = -\Lambda_{i,n_i} v_{i,n_i}^2 - \underline{\Lambda}_{i,n_i-1} v_{i,n_i-1} v_{i,n_i} + v_{i,n_i} \left[f_{i,n_i}^{\sigma_i(t)}(x_i) + g_{i,n_i}^{\sigma_i(t)}\left(x_i\left(t - \tau_{i,n_i}^{\sigma_i(t)}\right)\right) \right]$$

$$\begin{aligned}
& + \left(\bar{\rho}_{i,n_i} - 1 \right) \rho_{i,n_i} v_{i,n_i} \lambda_{i,n_i} - \frac{\Theta_{i,n_i}}{2\eta_{i,n_i}^2} v_{i,n_i}^2 \phi_{i,n_i}^T \phi_{i,n_i} + \Gamma_{i,n_i} \tilde{\Theta}_{i,n_i} \hat{\Theta}_{i,n_i} \\
& - \varpi_i \left(1 - \bar{\tau}_{i,n_i}^{\sigma_i(t)} \right) \check{V}_{i,n_i}^{\sigma_i(t)} + e^{\varpi_i \bar{\tau}_{i,n_i}^{\sigma_i(t)}} \left[g_{i,n_i}^{\sigma_i(t)} \left(x_i(t) \right) \right]^2 - \left(1 - \bar{\tau}_{i,n_i}^{\sigma_i(t)} \right) \left[g_{i,n_i}^{\sigma_i(t)} \left(x_i \left(t - \bar{\tau}_{i,n_i}^{\sigma_i(t)} \right) \right) \right]^2.
\end{aligned} \tag{52}$$

With [Assumption 3](#) and Young's inequality, we can obtain

$$\frac{1}{4(1 - \bar{\tau}_{d,i})} v_{i,n_i}^2 \geq v_{i,n_i} g_{i,n_i}^{\sigma_i(t)} \left(x_i \left(t - \bar{\tau}_{i,n_i}^{\sigma_i(t)} \right) \right) - (1 - \bar{\tau}_{d,i}) \left[g_{i,n_i}^{\sigma_i(t)} \left(x_i \left(t - \bar{\tau}_{i,n_i}^{\sigma_i(t)} \right) \right) \right]^2,$$

Combined with the above fact, we have

$$\begin{aligned}
\dot{V}_{i,n_i}^{\sigma_i(t)} & \leq -\Lambda_{i,n_i} v_{i,n_i}^2 - \underline{\Lambda}_{i,n_i-1} v_{i,n_i-1} v_{i,n_i} - \varpi_i (1 - \bar{\tau}_{d,i}) \check{V}_{i,n_i}^{\sigma_i(t)} + v_{i,n_i} H_{i,n_i}^{\sigma_i(t)} (x_i) \\
& - \frac{\Theta_{i,n_i}}{2\eta_{i,n_i}^2} v_{i,n_i}^2 \phi_{i,n_i}^T \phi_{i,n_i} + \Gamma_{i,n_i} \tilde{\Theta}_{i,n_i} \hat{\Theta}_{i,n_i} + \left[1 - 2 \tanh^2 \left(\frac{v_{i,n_i}}{\chi_{i,n_i}} \right) \right] G_{i,n_i}^{\sigma_i(t)} (x_i).
\end{aligned} \tag{53}$$

Applying the Young's inequality to (53) yields

$$\begin{aligned}
v_{i,n_i} H_{i,n_i}^{\sigma_i(t)} (x_i) & \leq \frac{v_{i,n_i}^2}{2\eta_{i,n_i}} \Theta_{i,n_i} \phi_{i,n_i}^T \phi_{i,n_i} + \frac{\eta_{i,n_i} + v_{i,n_i}^2 + \varepsilon_{i,n_i}^2}{2}, \\
\tilde{\Theta}_{i,n_i} \hat{\Theta}_{i,n_i} & \leq \frac{1}{2} \Theta_{i,n_i}^2 - \frac{1}{2} \tilde{\Theta}_{i,n_i}^2.
\end{aligned} \tag{54}$$

Substituting (54) into (53), we have

$$\begin{aligned}
\dot{V}_{i,n_i}^{\sigma_i(t)} & \leq -\left(\Lambda_{i,n_i} - \frac{1}{2} \right) v_{i,n_i}^2 - \underline{\Lambda}_{i,n_i-1} v_{i,n_i-1} v_{i,n_i} - \frac{\Gamma_{i,n_i}}{2} \tilde{\Theta}_{i,n_i}^2 - \varpi_i (1 - \bar{\tau}_{d,i}) \check{V}_{i,n_i}^{\sigma_i(t)} \\
& + \frac{\Gamma_{i,n_i} \Theta_{i,n_i}^2 + \eta_{i,n_i} + \varepsilon_{i,n_i}^2}{2} + \left[1 - 2 \tanh^2 \left(\frac{v_{i,n_i}}{\chi_{i,n_i}} \right) \right] G_{i,n_i}^{\sigma_i(t)} (x_i).
\end{aligned} \tag{55}$$

4. Stability analysis under asynchronous switching

4.1. Boundedness of the state and NN weight errors

The following lemma is required for the subsequent stability analysis.

Lemma 3. [54] Define a set $\Omega_v = \{v \mid |v| < 0.8814\chi\}$ with $\chi > 0$. Then, one has $1 - 2 \tanh^2 \left(\frac{v}{\chi} \right) \leq 0$ for all $v \notin \Omega_v$.

Accordingly, the stability analysis for the overall MASs with the presented design can be summarized as follows.

Theorem 1. Under [Assumption 1, 2, 3](#), for MASs with followers governed by (1) and the presented design in [Sections 3.2.1, 3.2.2, 3.2.3](#), the signals $v_{i,l}$, $\tilde{\Theta}_{i,l}$, and $x_{i,l}$ are UUB, for all i and l .

Proof. First, we consider the boundedness of the closed-loop signals on an interval $[t_{q_i,k}, t_{q_i,k+1})$ in which the i -th follower operates under a fixed mode $\sigma_i(t)$. Then, we discuss the effect of asynchronous switching during the interval $[0, T]$ with the overall Lyapunov-Krasovskii functional and the ADT condition in [Assumption 2](#).

1) Consider the Lyapunov-Krasovskii candidate as $V = \sum_{i=1}^N V_i^{\sigma_i(t)} = \sum_{i=1}^N \sum_{l=1}^{n_i} V_{i,l}^{\sigma_i(t)}$ for $t \in [t_{q_i,k}, t_{q_i,k+1})$. From (23), (38), and (55), we can obtain

$$\begin{aligned}
\dot{V} & = \sum_{i=1}^N \sum_{l=1}^{n_i} \dot{V}_{i,l}^{\sigma_i(t)} \\
& \leq -\sum_{i=1}^N \sum_{l=1}^{n_i} (2\Lambda_{i,l} - 1) \frac{v_{i,l}^2}{2} - \sum_{i=1}^N \sum_{l=1}^{n_i} \frac{\Gamma_{i,l} \tilde{\Theta}_{i,l}^2}{2} - \sum_{i=1}^N \left[\varpi_i (1 - \bar{\tau}_{d,i}) \sum_{l=1}^{n_i} \hat{V}_{i,l}^{\sigma_i(t)} \right] \\
& - \sum_{i=1}^N \sum_{j \in N_i} \varpi_j (1 - \bar{\tau}_{d,j}) \hat{V}_{j,1}^{\sigma_j(t)} + \sum_{i=1}^N \sum_{l=1}^{n_i} \frac{\Gamma_{i,l} \Theta_{i,l}^2 + \eta_{i,l} + \varepsilon_{i,l}^2}{2} + \sum_{i=1}^N \sum_{l=1}^{n_i} \left[1 - 2 \tanh^2 \left(\frac{v_{i,l}}{\chi_{i,l}} \right) \right] G_{i,l}^{\sigma_i(t)} \\
& \leq -aV + b + \sum_{i=1}^N \sum_{l=1}^{n_i} \left[1 - 2 \tanh^2 \left(\frac{v_{i,l}}{\chi_{i,l}} \right) \right] G_{i,l}^{\sigma_i(t)},
\end{aligned} \tag{56}$$

where

$$a = \min_{\substack{l=1, \dots, n_i \\ i=1, \dots, N}} \{2\Lambda_{i,l} - 1, \Gamma_{i,l}, \varpi_i (1 - \bar{\tau}_{d,i})\}, \quad b = \sum_{i=1}^N \sum_{l=1}^{n_i} \frac{\Gamma_{i,l} \Theta_{i,l}^2 + \eta_{i,l} + \varepsilon_{i,l}^2}{2}.$$

For the term $\sum_{i=1}^N \sum_{j=1}^{n_i} \left[1 - 2 \tanh^2 \left(\frac{v_{i,j}}{\chi_{i,j}} \right) \right] G_{i,j}^{\sigma_i(t)}$, we consider the following three cases. To begin with, define

$$\Omega_{i,j}^\vee = \left\{ v_{i,j} \mid |v_{i,j}| \leq 0.8814 \chi_{i,j} \right\}, \quad D = \left\{ (i,j) \mid v_{i,j} \in \Omega_{i,j}^\vee \right\},$$

$$\bar{\Omega}_{i,j}^\vee = \left\{ v_{i,j} \mid |v_{i,j}| > 0.8814 \chi_{i,j} \right\}, \quad \bar{D} = \left\{ (i,j) \mid v_{i,j} \notin \Omega_{i,j}^\vee \right\}.$$

Case 1 ($v_{i,j} \in \Omega_{i,j}^\vee$). In this case, $|v_{i,j}| \leq 0.8814 \chi_{i,j}$ holds for all $(i,j) \in D$ with a constant $\chi_{i,j} > 0$. Therefore, the compensating error $v_{i,j}$ is upper-bounded where the bound $0.8814 \chi_{i,j}$ can be made arbitrarily small.

Case 2 ($v_{i,j} \in \bar{\Omega}_{i,j}^\vee$). In this case, according to Lemma 3, we have $1 - 2 \tanh^2 \left(\frac{v_{i,j}}{\chi_{i,j}} \right) < 0$ for all $(i,j) \in \bar{D}$. Then, $\left[1 - 2 \tanh^2 \left(\frac{v_{i,j}}{\chi_{i,j}} \right) \right] G_{i,j}^{\sigma_i(t)} \leq 0$ because $G_{i,j} > 0$. Hence, $\dot{V} \leq -aV + b$, which indicates the ultimate boundedness of $v_{i,j}$ and $\bar{\Theta}_{i,j}$.

Case 3 ($v_{i,j} \in \Omega_{i,j}^\vee$ for $(i,j) \in D$ and $v_{i,j} \in \bar{\Omega}_{i,j}^\vee$ for $(i,j) \in \bar{D}$). For $(i,j) \in D$, we can conclude that $|v_{i,j}| \leq 0.8814 \chi_{i,j}$, which follows from Case 1. For $(i,j) \in \bar{D}$, we can obtain $\left[1 - 2 \tanh^2 \left(\frac{v_{i,j}}{\chi_{i,j}} \right) \right] G_{i,j} \leq 0$, which follows from Case 2.

To summarize, there exists $\bar{b} \geq b$ such that

$$\dot{V} \leq -aV + \bar{b}$$

which implies that the terms $v_{i,j}$, $\bar{\Theta}_{i,j}$ are ultimate bounded for $t \in [t_{q_{i,k}}, t_{q_{i,k+1}})$, $i = 1, \dots, N$, $j = 1, \dots, n_i$.

2) Next, we investigate the boundedness of all signals in the interval $[0, T]$. From the previous discussions, it was verified that there exists a constant $\Phi_i > 1$, a_i and $\bar{b}_i$ such that

$$V_i^q \leq \Phi_i V_i^p, \forall p, q \in \mathcal{Q}_i, \quad \dot{V}_i^{\sigma_i(t)} \leq -a_i V_i^{\sigma_i(t)} + \bar{b}_i. \quad (57)$$

Define $\bar{V}_i(t) = e^{\kappa_i t} V_i^{\sigma_i(t)}(t)$. Since $\bar{V}_i(t)$ is piecewise differentiable in each interval $[t_{q_{i,k}}, t_{q_{i,k+1}})$, we have

$$\dot{\bar{V}}_i = \kappa_i \bar{V}_i + e^{\kappa_i t} \dot{V}_i^{\sigma_i(t)} \leq \bar{b}_i e^{\kappa_i t}. \quad (58)$$

Combining (57) and (58) yields

$$\bar{V}_i(t_{q_{i,k+1}}) = e^{\kappa_i t_{q_{i,k+1}}} V_i^{\sigma_i(t_{q_{i,k+1}})}(t_{q_{i,k+1}}) \leq \Phi_i \bar{V}_i(t_{q_{i,k+1}}^-), \quad (59)$$

where $\bar{V}_i(t_{q_{i,k+1}}^-) = e^{\kappa_i t_{q_{i,k+1}}} V_i^{\sigma_i(t_{q_{i,k+1}})}(t_{q_{i,k+1}}^-)$. Based on (58), we have

$$\bar{V}_i(t_{q_{i,k+1}}) \leq \bar{V}_i(t_{q_{i,k+1}}^-) \leq \Phi_i \left[\bar{V}_i(t_{q_{i,k}}) + \int_{t_{q_{i,k}}}^{t_{q_{i,k+1}}} \bar{b}_i e^{\kappa_i t} dt \right]. \quad (60)$$

Integrating (60) yields

$$\bar{V}_i(T^-) = \bar{V}_i(t_{q_{i,\bar{N}}}) + \int_{t_{q_{i,\bar{N}}}}^T \bar{b}_i e^{\kappa_i t} dt, \quad (61)$$

where $\bar{N} = N_{\sigma_i}(T, 0)$. Recalling (58), we have

$$\bar{V}_i(t_{q_{i,\bar{N}}}) \leq \Phi_i \left[\bar{V}_i(t_{q_{i,\bar{N}-1}}) + \int_{t_{q_{i,\bar{N}-1}}}^{t_{q_{i,\bar{N}}}} \bar{b}_i e^{\kappa_i t} dt \right] \leq \Phi_i^2 \left[\bar{V}_i(t_{q_{i,\bar{N}-2}}) + \int_{t_{q_{i,\bar{N}-2}}}^{t_{q_{i,\bar{N}-1}}} \bar{b}_i e^{\kappa_i t} dt \right] + \Phi_i \int_{t_{q_{i,\bar{N}-1}}}^{t_{q_{i,\bar{N}}}} \bar{b}_i e^{\kappa_i t} dt. \quad (62)$$

Similarly, by recursion, we have

$$\bar{V}_i(t_{N_{\sigma_i}(T,0)}) \leq (\Phi_i)^{\bar{N}} \bar{V}_i(0) + \sum_{k=0}^{\bar{N}-1} (\Phi_i)^{\bar{N}-k} \int_{t_{q_{i,k}}}^{t_{q_{i,k+1}}} \bar{b}_i e^{\kappa_i t} dt. \quad (63)$$

Collecting the results in the above equations, we have

$$V_{i,W}(T^-) \leq \Phi_i^{N_{\sigma_i}(T,0)} V_{i,W}(0) + \int_{N_{\sigma_i}(T,0)}^T \ell_i' e^{\kappa_i t} dt + \sum_{k_i=0}^{N_{\sigma_i}(T,0)-1} \Phi_i^{N_{\sigma_i}(T,0)-k_i} \int_{t_{k_i}}^{t_{k_i+1}} \ell_i' e^{\kappa_i t} dt. \quad (64)$$

Moreover, suppose that $\tau_{a,i} > \frac{\log \Phi_i}{i} \kappa_i$, there exists a constant r_i such that $0 < r_i < \kappa_i - \frac{\log \Phi_i}{\tau_{a,i}}$. From Definition 1 and (64), we have

$$N_{\sigma_i}(T, 0) - q_{i,k} \leq 1 + N_{\sigma_i}(T, t_{q_{i,k+1}}) \leq 1 + N_0 + \frac{T - t_{q_{i,k+1}}}{\tau_{a,i}} \leq 1 + N_0 + \frac{(\kappa_i - r_i)(T - t_{q_{i,k+1}})}{\log \Phi_i} \quad (65)$$

for $q_{i,k} = 0, 1, \dots, N_{\sigma_i}(T, 0)$. Due to $r_i < \kappa_i$, one has

$$\int_{t_{k_i}}^{t_{k_i+1}} e^{\kappa_i t} dt \leq e^{(\kappa_i - r_i)t_{k_i+1}} \int_{t_{k_i}}^{t_{k_i+1}} e^{r_i t} dt. \quad (66)$$

Inserting (65) and (66) into (64) yields

$$\bar{V}_i(T^-) \leq (\Phi_i)^{\bar{N}} \bar{V}_i(0) + (\Phi_i)^{1+N_0} e^{(\kappa_i - r_i)T} \int_0^T \bar{b}_i e^{r_i t} dt. \quad (67)$$

From the definition of $\bar{V}_i$ and the above fact, we have

$$V_i^{\sigma_i(T^-)}(T^-) \leq e^{-\kappa_i T} (\Phi_i)^{\bar{N}} \bar{V}_i(0) + (\Phi_i)^{1+N_0} \frac{\bar{b}_i}{r_i} (1 - e^{-r_i T}), \quad (68)$$

which implies that $V_i^{\sigma_i(T)}$ is bounded for $t \in [0, T]$ and satisfies

$$\lim_{T \rightarrow +\infty} V_i^{\sigma_i(T)}(T) \leq (\Phi_i)^{1+N_0} \frac{\bar{b}_i}{r_i}. \quad (69)$$

Therefore, the ultimate boundedness of $v_{i,j}$ and $\tilde{\Theta}_{i,j}$ can be ensured. This completes the proof. $\square$

Remark 8. The inequalities (67)–(69) in Theorem 1 show that the global Lyapunov function decreases exponentially up to a constant offset, which leads directly to UUB of all closed-loop signals. In particular, the bipartite synchronization errors converge to a small neighborhood of the origin whose size can be made arbitrarily small by tuning the design parameters and RBFNN approximation accuracy.

Remark 9. The ADT condition in Assumption 2 ensures that each switching mode is active for a sufficiently long time interval, on average, so that the compensation filters and adaptive laws have enough time to attenuate the corresponding error signals. In particular, the dynamics of $\beta_{i,l}$ and $\lambda_{i,l}$ are mode-dependent, and the negative terms in the Lyapunov-Krasovskii functionals rely on the convergence of these filter states. The ADT bound guarantees that the cumulative decrease of the Lyapunov function during the dwell intervals dominates any possible increase at the switching instants, which is essential for establishing uniform ultimate boundedness of the closed-loop signals under asynchronous switching.

Remark 10. The design parameters $\Gamma_{i,l}$, $\Lambda_{i,l}$, $\eta_{i,l}$ are, in principle, allowed to depend on both the agent index i and the layer l . The stability analysis only requires that they satisfy the inequalities stated in Theorem 1, such as $\Lambda_{i,l} > \frac{1}{2}$, $\Gamma_{i,l}, \eta_{i,l} > 0$ to ensure $a > 0$ in (56). In practical implementations, one may either select agent-dependent gains to account for heterogeneous dynamics, or choose uniform gains to simplify the tuning process.

4.2. Boundedness of closed-loop signals

It should be noted that the boundedness of $v_{i,l}$ does not directly imply the boundedness of $\delta_{i,l}$, $\xi_{i,l}$, $\tilde{\alpha}_{i,l}$ and $\beta_{i,l}$, which is discussed further in this subsection.

Corollary 1. For each follower, the DSC filter error $\tilde{\alpha}_{i,l}$ and compensating signal $\beta_{i,l}$ are bounded.

Proof. From the error signals defined in (11) and filter designs in (5), we have

$$\dot{\tilde{\alpha}}_{i,l} = \dot{\alpha}_{i,l} - \dot{s}_{i,l} = -\frac{\tilde{\alpha}_{i,l}}{\omega_{i,l}} + B_{i,l}, \quad j = 1, \dots, n_i - 1$$

with

$$B_{i,1} = \frac{1}{o_i} (\Lambda_{i,1} \dot{\delta}_{i,1} - b_i \ddot{x}_0 + o_i \dot{\lambda}_{i,2}) + \frac{1}{o_i} \left(\frac{\dot{\hat{\Theta}}_{i,1}}{2\eta_{i,1}} v_{i,1} \phi_{i,1}^T \phi_{i,1} + \frac{\hat{\Theta}_{i,1}}{2\eta_{i,1}} \dot{v}_{i,1} \phi_{i,1}^T \phi_{i,1} \right) + \frac{1}{o_i} \left(\frac{\hat{\Theta}_{i,1}}{\eta_{i,1}} v_{i,1} \phi_{i,1}^T \dot{\phi}_{i,1} - \sum_{j \in \mathcal{N}_i} \kappa_{i,j} \dot{x}_{j,2} \right)$$

$$B_{i,l} = \frac{\Lambda_{i,l-1} \dot{\delta}_{i,l-1} - \Lambda_{i,l} \dot{\delta}_{i,l}}{2\eta_{i,l}} + \frac{\dot{\hat{\Theta}}_{i,l}}{2\eta_{i,l}} v_{i,l} \phi_{i,l}^T \phi_{i,l} + \frac{\hat{\Theta}_{i,l}}{2\eta_{i,l}} \dot{v}_{i,l} \phi_{i,l}^T \phi_{i,l} + \frac{\hat{\Theta}_{i,l}}{\eta_{i,l}} v_{i,l} \phi_{i,l}^T \dot{\phi}_{i,l} + \rho_{i,l} \dot{\lambda}_{i,l} - \tilde{\alpha}_{i,l-1}, \quad j = 2, \dots, n_i - 1,$$

being continuous on the compact sets $\Pi_{x_0} \times \Pi_{i,l}$, which are defined as

$$\Pi_{x_0} = \{ (x_0, \dot{x}_0, \ddot{x}_0) : x_0^2 + \dot{x}_0^2 + \ddot{x}_0^2 \leq R_0 \},$$

$$\Pi_{i,l} = \left\{ (\delta_{i,l}, \tilde{\alpha}_{i,l}, \tilde{\Theta}_{i,l}) : \frac{1}{2} \delta_{i,l}^2 + \frac{1}{2} \tilde{\alpha}_{i,l}^2 + \frac{1}{2} \tilde{\Theta}_{i,l}^2 \leq R_{i,l} \right\},$$

with positive constants R_0 and $R_{i,l}$. Consequently, there exists a positive constant $M_{i,l} > 0$ such that $|B_{i,l}| \leq M_{i,l}$. Applying Young's inequality yields

$$B_{i,l} \tilde{\alpha}_{i,l} \leq M_{i,l} |\tilde{\alpha}_{i,l}| \leq \frac{M_{i,l}^2}{2h_{i,l}} \tilde{\alpha}_{i,l}^2 + \frac{h_{i,l}}{2}, \quad (70)$$

with $h_{i,l} > 0$.

Consider the Lyapunov candidate for $\tilde{\alpha}_{i,l}$ and $\beta_{i,l}$ as

$$V_{\alpha\beta} = \frac{1}{2} \sum_{j=1}^{n_i-1} \tilde{\alpha}_{i,l}^2 + \frac{1}{2} \sum_{j=1}^{n_i} \beta_{i,l}^2.$$

with

$$\dot{V}_{\alpha\beta} = \sum_{j=1}^{n_i-1} \tilde{\alpha}_{i,l} \dot{\tilde{\alpha}}_{i,l} + \sum_{j=1}^{n_i} \beta_{i,l} \dot{\beta}_{i,l}.$$

Given (6), (7), (8), and (70), we can obtain

$$\dot{V}_{\alpha\beta} \leq - \sum_{j=1}^{n_i-1} \left(\frac{1}{\omega_{i,l}} - \frac{M_{i,l}^2}{2h_{i,l}} \right) \tilde{\alpha}_{i,l}^2 + \sum_{j=1}^{n_i-1} \frac{h_{i,l}}{2} - \sum_{j=1}^{n_i} \Lambda_{i,l} \beta_{i,l}^2 + \sum_{j=1}^{n_i-1} \Delta_{i,l} \tilde{\alpha}_{i,l} \beta_{i,l}. \quad (71)$$

The term $\sum_{j=1}^{n_i-1} \Delta_{i,l} \tilde{\alpha}_{i,l} \beta_{i,l}$ in (71) satisfies

$$\sum_{j=1}^{n_i-1} \Delta_{i,l} \tilde{\alpha}_{i,l} \beta_{i,l} \leq \sum_{j=1}^{n_i-1} \frac{\Delta_{i,l}}{2} \tilde{\alpha}_{i,l}^2 + \sum_{j=1}^{n_i-1} \frac{\Delta_{i,l}}{2} \beta_{i,l}^2. \quad (72)$$

Define $\tilde{\omega}_{i,l} = \frac{1}{\omega_{i,l}} - \frac{M_{i,l}^2}{2h_{i,l}} - \frac{\Delta_{i,l}}{2}$ for $l = 1, \dots, n_i - 1$. Inserting (72) into (71) yields

$$\dot{V}_{\alpha\beta} \leq - \sum_{j=1}^{n_i-1} \tilde{\omega}_{i,l} \tilde{\alpha}_{i,l}^2 + \sum_{j=1}^{n_i-1} \frac{h_{i,l}}{2} - \sum_{j=1}^{n_i} \left(\Lambda_{i,l} - \frac{\Delta_{i,l}}{2} \right) \beta_{i,l}^2. \quad (73)$$

Then, the fact in (73) implies the boundedness of the signals $\tilde{\alpha}_{i,l}$ and $\beta_{i,l}$. This completes the proof. $\square$

Corollary 2. For each follower, the distributed signal $\delta_{i,l}$ is bounded.

Proof. From the error definition, the fact that $v_{i,l} = \delta_{i,l} - \beta_{i,l}$ implies that $\delta_{i,l} = v_{i,l} + \beta_{i,l}$. Theorem 1 implies the boundedness of $v_{i,l}$ and Corollary 1 validates the boundedness of $\beta_{i,l}$. Thus, the error signal $\delta_{i,l}$ is bounded. This completes the proof. $\square$

Corollary 3. For each follower, the distributed dead-zone compensating filter $\lambda_{i,l}$ is bounded if the design parameters satisfy $\bar{\rho}_{i,n_i} > d_i + 1$ and $\rho_{i,n_i} > \frac{1}{(\bar{\rho}_{i,n_i} - d_i - 1)}$.

Proof. First, from Theorem 1, $\tilde{\Theta}$ and v are bounded, which implies that $\frac{\tilde{\Theta}_{i,n_i}}{2\eta_{i,n_i}} v_{i,n_i} \phi_{i,n_i}^T \phi_{i,n_i}$ is also bounded. Moreover, the boundedness of $\tilde{\alpha}_{i,l}$ and $\delta_{i,l}$ can be obtained based on Corollaries 1 and 2, respectively. Given these facts, there exists a positive constant U_i such that $\left\| \frac{\tilde{\Theta}_{i,n_i}}{2\eta_{i,n_i}} v_{i,n_i} \phi_{i,n_i}^T \phi_{i,n_i} + \frac{\tilde{\alpha}_{i,l}}{\omega_{i,l}} \right\| \leq U_i$. Therefore, the control design (45) satisfies

$$\|u_i\| \leq U_i + \rho_{i,n_i} \|\lambda_{i,n_i}\|. \quad (74)$$

With the dead-zone mapping properties in (2) and (3), we have

$$\|\Psi_i(u_i)\| \leq d_i \|u_i\| + c_i \leq d_i \rho_{i,n_i} \|\lambda_{i,n_i}\| + d_i U_i + c_i. \quad (75)$$

Accordingly, the compensating signal λ_{i,n_i} satisfies

$$\begin{aligned} \lambda_{i,n_i} \dot{\lambda}_{i,n_i} &= \lambda_{i,n_i} \left[-\bar{\rho}_{i,n_i} \rho_{i,n_i} \lambda_{i,n_i} + \Psi_i(u_i) - u_i \right] \\ &\leq \left\| \lambda_{i,n_i} \right\| \left(d_i \rho_{i,n_i} \|\lambda_{i,n_i}\| + \rho_{i,n_i} \|\lambda_{i,n_i}\| + d_i U_i + c_i + U_i \right) - \bar{\rho}_{i,n_i} \rho_{i,n_i} \lambda_{i,n_i}^2 \\ &\leq (d_i + 1 - \bar{\rho}_{i,n_i}) \rho_{i,n_i} \lambda_{i,n_i}^2 + \frac{\lambda_{i,n_i}^2}{2} + \frac{[(d_i + 1)U_i + c_i]^2}{2}. \end{aligned}$$

Finally, differentiating the Lyapunov candidate $V_{i,\lambda} = \frac{1}{2} \sum_{j=1}^{n_i} \lambda_{i,j}^2$ yields

$$\begin{aligned} \dot{V}_{i,\lambda} &= \sum_{j=1}^{n_i-1} \lambda_{i,j} \dot{\lambda}_{i,j+1} - \sum_{j=1}^{n_i-1} \rho_{i,j} \lambda_{i,j}^2 + \lambda_{i,n_i} \dot{\lambda}_{i,n_i} \\ &\leq -\left(\rho_{i,1} - \frac{1}{2} \right) \frac{\lambda_{i,1}^2}{2} - \sum_{l=2}^{n_i-1} (\rho_{i,l} - 1) \lambda_{i,l}^2 - (\hat{\rho}_{i,n_i} - 1) \lambda_{i,n_i}^2 + \frac{[(d_i + 1)U_i + c_i]^2}{2}, \end{aligned} \quad (76)$$

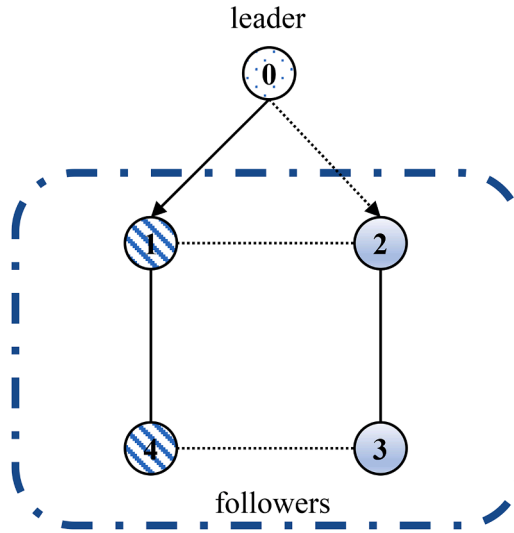


Fig. 1. Communication topology.

with $\hat{\rho}_{i,n_i} = (\bar{\rho}_{i,n_i} - d_i - 1)\rho_{i,n_i}$. Therefore, we can obtain $\dot{V}_{i,\lambda} \leq c_{i,\lambda}V_{i,\lambda} + d_{i,\lambda}$ with

$$c_{i,\lambda} = \max_{l=2,\dots,n_i-1} \left\{ \rho_{i,1} - \frac{1}{2}, \rho_{i,l} - 1, \hat{\rho}_{i,n_i} - 1 \right\}, \quad d_{i,\lambda} = \frac{[(d_i + 1)U_i + c_i]^2}{2},$$

which implies the boundedness of the signals $\lambda_{i,l}$ for $l = 1, \dots, n_i$. This completes the proof. $\square$

Corollary 4. *The distributed signals $\xi_{i,l}$ are bounded.*

Proof. First, the error definition $\delta_{i,l} = \xi_{i,l} - \lambda_{i,l}$ implies that $\xi_{i,l} = \delta_{i,l} + \lambda_{i,l}$. Next, the boundedness of the signals $\delta_{i,l}$ and $\lambda_{i,l}$ can be obtained based on Corollaries 2 and 3, respectively. Therefore, the distributed signals $\xi_{i,l}$ are bounded for all $i = 1, \dots, N$ and $l = 1, \dots, n_i - 1$. This completes the proof. $\square$

It is worth emphasizing that the UUB result in Theorem 1 and Corollaries 1, 2, 3, 4 is intrinsic to the considered nonlinear switched MASs with time-varying state delays, dead-zone nonlinearities, and NN approximation. The delay-dependent terms associated with Assumption 3 and the bounded approximation residuals $\varepsilon_{i,l}^{(i)}(\cdot)$ give rise to additive positive constants in the Lyapunov derivatives, which prevent establishing $\dot{V} \leq -cV$ for some $c > 0$. If exact models without approximation residuals and stronger delay conditions were imposed, one could in principle recover asymptotic convergence of $\xi_{i,1}$ by refining the same Lyapunov-Krasovskii framework. A rigorous treatment of such strengthened conditions is left for future work.

From an implementation viewpoint, the proposed protocol scales favorably with respect to the network size, since each agent only requires local information from its neighbors and the leader. The computational burden at each node is dominated by the neural network evaluation and the update of a small number of filter and adaptive states, and it grows essentially linearly with the node degree. On the other hand, the ADT condition provides an explicit constraint on the admissible switching frequency: if the switching becomes excessively fast, the stability guarantees are no longer valid. This suggests that, in practice, the mode-switching mechanism should enforce a minimal ADT, which is compatible with physical and communication limits in large-scale MASs.

5. Simulation example

In this section, a group of MASs with a leader labeled as 0 and followers labeled as 1 to 4 is considered with the communication topology displayed in Fig. 1.

5.1. Numerical validation

Here, the solid line represents $a_{i,j} = 1$ or $b_i = 1$, while the dashed line suggests $a_{i,j} = -1$ or $b_i = -1$. The follower set $\mathcal{F} = \{1, 2, 3, 4\}$ is further divided into two sets $\mathcal{F}_1 = \{1, 4\}$ and $\mathcal{F}_2 = \{2, 3\}$. In addition, we can easily verify that the graph among followers is structurally balanced. The bipartite synchronization objective is to design distributed protocols such that $y_i \rightarrow x_0$ for $i \in \mathcal{F}_1$ and $y_i \rightarrow -x_0$ for $i \in \mathcal{F}_2$ as $t \rightarrow +\infty$.

The leader is considered as $x_0 = \sin(0.5t)$. For the followers, the heterogeneous dynamics are considered as $n_1 = n_3 = n_4 = 2, n_2 = 3$, with

$$\begin{cases} f_{1,1}^1(x_{1,1}) = 0.1x_{1,1}, g_{1,1}^1(x_{1,1}) = 0.1x_{1,1} \\ f_{1,2}^1(\bar{x}_{1,2}) = 0.3x_{1,1}x_{1,2}, g_{1,2}^1(\bar{x}_{1,2}) = 0.1 \\ \tau_{1,1}^1 = 0.15 \end{cases},$$

for follower 1 with $\sigma_1 = 1$,

$$\begin{cases} f_{1,1}^2(x_{1,1}) = -0.2x_{1,1}, g_{1,1}^2(x_{1,1}) = 0.15x_{1,1} \\ f_{1,2}^2(\bar{x}_{1,2}) = 0.1x_{1,1}x_{1,2}, g_{1,2}^2(\bar{x}_{1,2}) = 0.3x_{1,2} \\ \tau_{1,1}^2 = 0.1[\sin(t) + 1], \tau_{1,2}^2 = 0.12 \end{cases},$$

for follower 1 with $\sigma_1 = 2$,

$$\begin{cases} f_{1,1}^3(x_{1,1}) = 0.15x_{1,1}, g_{1,1}^3(x_{1,1}) = 0.3x_{1,1} \\ f_{1,2}^3(\bar{x}_{1,2}) = 0.2x_{1,1}x_{1,2}, g_{1,2}^3(\bar{x}_{1,2}) = 0.4 \\ \tau_{1,1}^3 = 0.14 \end{cases},$$

for follower 1 with $\sigma_1 = 3$,

$$\begin{cases} f_{2,1}^1(x_{2,1}) = -0.2x_{2,1}, g_{2,1}^1(x_{2,1}) = 0.15x_{2,1} \\ f_{2,2}^1(\bar{x}_{2,2}) = 0.1x_{2,1}x_{2,2}, g_{2,2}^1(\bar{x}_{2,2}) = 0.3x_{2,2} \\ f_{2,3}^1(\bar{x}_{2,3}) = 0.15x_{2,1}x_{2,2}x_{2,3}, g_{2,3}^1(\bar{x}_{2,3}) = 0.1x_{2,1}x_{2,3} \\ \tau_{2,1}^1 = 0.1[1 + \sin(t)], \tau_{2,2}^1 = 0.12 \\ \tau_{2,3}^1 = 0.1 \end{cases},$$

for follower 2 with $\sigma_2 = 1$,

$$\begin{cases} f_{2,1}^2(x_{2,1}) = -0.1x_{2,1}, g_{2,1}^2(x_{2,1}) = 0.1x_{2,1} \\ f_{2,2}^2(\bar{x}_{2,2}) = 0.15x_{2,1}x_{2,2}, g_{2,2}^2(\bar{x}_{2,2}) = 0.2x_{2,2} \\ f_{2,3}^2(\bar{x}_{2,3}) = 0.2x_{2,1}x_{2,2}x_{2,3}, g_{2,3}^2(\bar{x}_{2,3}) = 0.3x_{2,1}x_{2,3} \\ \tau_{2,1}^2 = 0.06[\sin(t) + 1], \tau_{2,2}^2 = 0.2 \\ \tau_{2,3}^2 = 0.2[0.3 + 0.1 \sin(t)] \end{cases},$$

for follower 2 $\sigma_2 = 2$,

$$\begin{cases} f_{3,1}^2(x_{3,1}) = -0.3x_{3,1}e^{-x_{3,1}}, g_{3,1}^2(x_{3,1}) = -0.3 \\ f_{3,2}^2(\bar{x}_{3,2}) = -0.1x_{3,2}, g_{3,2}^2(\bar{x}_{3,2}) = 0.2 \sin(x_{3,2}) \\ \tau_{3,2}^2 = 0.05[0.3 + 0.2 \cos(0.3t)] \end{cases},$$

for follower 3 with $\sigma_3 = 2$,

$$\begin{cases} f_{3,1}^3(x_{3,1}) = 0.3x_{3,1}, g_{3,1}^3(x_{3,1}) = -0.3 \\ f_{3,2}^3(\bar{x}_{3,2}) = 0.1x_{3,1}x_{3,2}, g_{3,2}^3(\bar{x}_{3,2}) = 0.15x_{3,1} \sin(x_{3,2}) \\ \tau_{3,2}^3 = 0.06[0.3 + 0.1 \sin(0.3t)] \end{cases},$$

for follower 3 with $\sigma_3 = 3$,

$$\begin{cases} f_{4,1}^1(x_{4,1}) = -0.1x_{4,1}e^{-x_{4,1}}, g_{4,1}^1(x_{4,1}) = -0.2 \\ f_{4,2}^1(\bar{x}_{4,2}) = -0.2x_{4,2}, g_{4,2}^1(\bar{x}_{4,2}) = 0.1 \sin(x_{4,2}) \\ \tau_{4,2}^1 = 0.04[0.2 + 0.1 \cos(0.3t)] \end{cases},$$

for follower 4 with $\sigma_4 = 1$,

$$\begin{cases} f_{4,1}^2(x_{4,1}) = 0.1x_{4,1}, g_{4,1}^2(x_{4,1}) = -0.2 \\ f_{4,2}^2(\bar{x}_{4,2}) = 0.2x_{4,1}x_{4,2}, g_{4,2}^2(\bar{x}_{4,2}) = 0.3x_{4,1} \sin(x_{4,2}) \\ \tau_{4,2}^2 = 0.1[1 + \sin(t)] \end{cases},$$

for follower 4 with $\sigma_4 = 2$,

$$\begin{cases} f_{4,1}^3(x_{4,1}) = 0.2x_{4,1}, g_{4,1}^3(x_{4,1}) = -0.5 \\ f_{4,2}^3(\bar{x}_{4,2}) = 0.3x_{4,1}x_{4,2}, g_{4,2}^3(\bar{x}_{4,2}) = 0.1x_{4,1} \sin(x_{4,2}) \\ \tau_{4,2}^3 = 0.08[0.5 + 0.4 \sin(0.3t)] \end{cases},$$

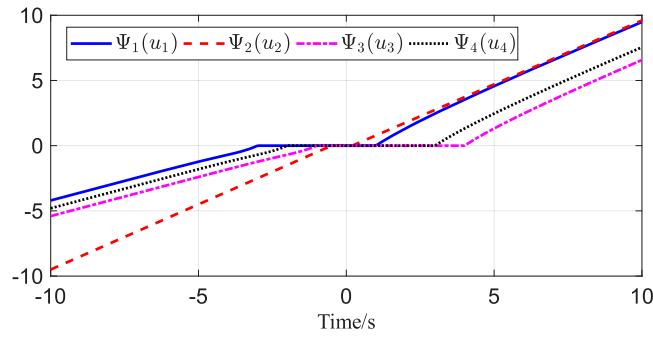


Fig. 2. The curves of input dead-zones Ψ_i of follower 1–4.

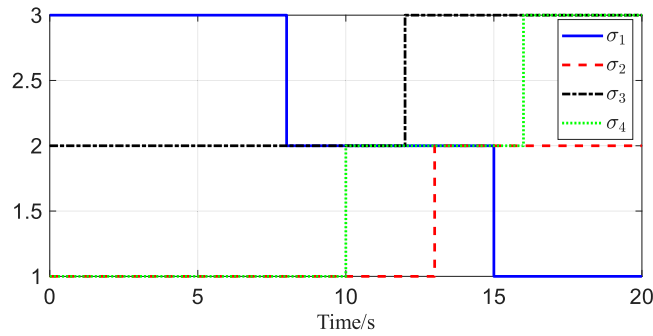


Fig. 3. The evolution of switching signals $\sigma_i(t)$ of follower 1–4.

for follower 4 with $\sigma_4 = 3$.

The input dead-zone nonlinearity is selected as

$$\Psi_1(u_1) = \begin{cases} [1 + 0.4 \tanh(u_1 + 1)](u_1 + 1), & u_1 \leq -1, \\ 0, & -1 < u_1 < 3, \\ 0.9(u_1 - 3) + 0.6 \log(u_1 - 2), & u_1 \geq 3 \end{cases}, \quad \Psi_2(u_2) = \begin{cases} u_2 + 0.2, & u_2 \leq -0.2, \\ 0, & -0.2 < u_2 < 0.5, \\ 0.98(u_2 - 0.5), & u_2 \geq 0.5 \end{cases},$$

$$\Psi_3(u_3) = \begin{cases} [1 + 0.4 \tanh(u_3 + 1)](u_3 + 1), & u_3 \leq -4, \\ 0, & -4 < u_3 < 1, \\ 0.9(u_3 - 1) + 0.6 \log(u_3), & u_3 \geq 1 \end{cases},$$

$$\Psi_4(u_4) = \begin{cases} [1 + 0.4 \tanh(u_4 + 1)](u_4 + 1), & u_4 \leq -3, \\ 0, & -3 < u_4 < 2, \\ 0.9(u_4 - 2) + 0.6 \log(u_4 - 1), & u_4 \geq 2, \end{cases}.$$

which are depicted in Fig. 2.

The centers of the RBFNNs are selected on a fixed, uniform grid over a compact set that contains the trajectories of the relevant state and error variables. Specifically, the RBFs $\phi_{i,l}(\cdot)$ are selected as Gaussian functions in Lemma 1, with the center spaced evenly in the region $[-2, 0, 2] \times [-2, 0, 2] \times \cdots \times [-2, 0, 2]$ and the width being set as 1.

The simulation is run with the parameters selected as $m_{1,1} = 34$, $m_{1,2} = 205$, $\omega_{1,1} = 0.01$, $\Gamma_{1,1} = 15$, $\Gamma_{1,2} = 26$, $\eta_{1,1} = 0.2$, $\eta_{1,2} = 0.25$, $\rho_{1,1} = 5$, $\rho_{1,2} = 2.5$, $\Delta_{1,\lambda} = 2$ for Follower 1, $m_{2,1} = 30$, $m_{2,2} = 208$, $m_{2,3} = 300$, $\omega_{2,1} = 0.01$, $\omega_{2,2} = 0.01$, $\Gamma_{2,1} = 10$, $\Gamma_{2,2} = 16$, $\Gamma_{2,3} = 28$, $\eta_{2,1} = 0.3$, $\eta_{2,2} = 0.5$, $\eta_{2,3} = 1$, $\rho_{2,1} = 6$, $\rho_{2,2} = 6$, $\rho_{2,3} = 2$, $\Delta_{2,\lambda} = 2.5$ for Follower 2, $m_{3,1} = 36$, $m_{3,2} = 211$, $\omega_{3,1} = 0.01$, $\Gamma_{3,1} = 10$, $\Gamma_{3,2} = 12$, $\eta_{3,1} = 0.8$, $\eta_{3,2} = 0.9$, $\rho_{3,1} = 5$, $\rho_{3,2} = 2$, $\Delta_{3,\lambda} = 3$, $\Delta_{4,\lambda} = 2$ for Follower 3, and $m_{4,1} = 30$, $m_{4,2} = 200$, $\omega_{4,1} = 0.01$, $\Gamma_{4,1} = 11$, $\Gamma_{4,2} = 10$, $\eta_{4,1} = 0.2$, $\eta_{4,2} = 0.5$, $\rho_{4,1} = 6$, $\rho_{4,2} = 2.5$ for Follower 4.

The simulation results are presented in Figs. 3–5. Fig. 3 provides the evolution of all switching signals. In Fig. 4, the evolution of the control signals is depicted. Fig. 5 displays the evolution of the leader and all followers' states, which indicates that bipartite synchronization is achieved with the presented protocol design.

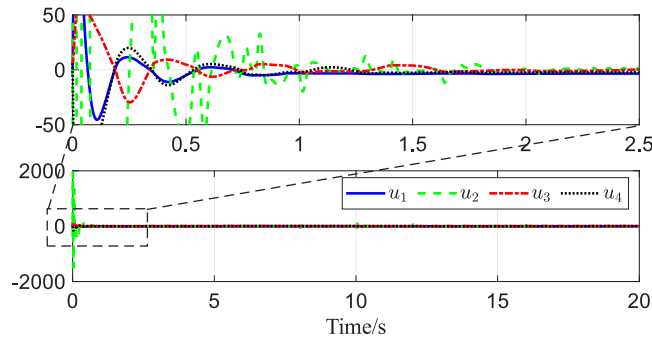


Fig. 4. The evolution of the input signals u_i with a zoomed-in plot.

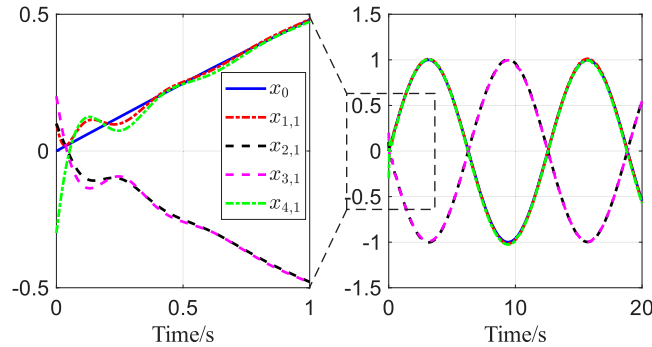


Fig. 5. The state trajectories of the leader and followers 1–4 with a zoomed-in plot.

5.2. Quantitative effect of the ADT

According to Definition 1, if a switching signal $\sigma(t)$ has an average dwell time $\mathcal{T}_a$ and a chatter bound $\bar{N}$, then for any interval $[t_0, t_f]$ the number of switches satisfies

$$N_\sigma(t_0, t_f) \leq \bar{N} + \frac{t_f - t_0}{\mathcal{T}_a}, \quad (77)$$

where $N_\sigma(t_0, t_f)$ denotes the number of discontinuity points of $\sigma(t)$ on (t_0, t_f) . In our simulation, the switching signals $\sigma_i(t)$ of all followers (see Fig. 3) are generated over the horizon $[0, 20]$ s such that they admit a common ADT $\mathcal{T}_{a,i} = \mathcal{T}_a$ and chatter bound $\bar{N}_i = \bar{N}$. Consequently, by (77) one obtains the upper bound

$$N_{\sigma_i}(0, 20) \leq \bar{N}_i + \frac{20}{\mathcal{T}_{a,i}}, \quad i = 1, \dots, N, \quad (78)$$

which directly limits the admissible switching frequency of each follower to

$$f_{sw,i} := \frac{N_{\sigma_i}(0, 20)}{20} \leq \frac{1}{\mathcal{T}_{a,i}} + \frac{\bar{N}_i}{20}. \quad (79)$$

From (78) and (79), one can observe that choosing a smaller $\mathcal{T}_{a,i}$ (or a larger chatter bound $\bar{N}_i$) allows more frequent switching, i.e., a larger admissible $f_{sw,i}$. One can verify from Fig. 3 that the actual number of switches $N_{\sigma_i}(0, 20)$ for each follower corresponds to the ADT, which is consistent with above analysis from (77) to (79).

Beyond the numerical example presented in this paper, the proposed distributed protocol and two-layer compensation filter can be readily extended to practical multi-agent applications such as UAV swarm coordination and mobile robot formation [55–57]. In such systems, switching among flight or operation modes naturally induces asynchronous switching, actuator dead-bands and static friction can be modeled as dead-zone input nonlinearities, and sensing and communication imperfections give rise to time-varying state delays. Because the controller design relies only on local neighbor information over the signed interaction graph and does not require any application-specific hardware structure, the same framework can be implemented on UAVs or mobile robots with minimal modification of the controller architecture. Experimental validation of these extensions will be pursued in future work.

6. Conclusion

In this paper, bipartite synchronization has been investigated for heterogeneous switched nonlinear MASs with time-varying state delays and input dead-zone nonlinearities. An adaptive distributed protocol combining RBFNNs, command filters, and a two-layer

compensation mechanism was proposed to deal with unknown dynamics. By employing the Lyapunov-Krasovskii framework with ADT that accommodates multiple time-varying delays and asynchronous switching, UUB of all closed-loop signals and practical bipartite synchronization were established and illustrated by a numerical example.

The results rely on bounded time-varying delays with bounded derivatives, and full-state measurements, which may be restrictive for networks with stochastic delays or more general actuator constraints. Future work will therefore focus on relaxing the delay assumptions, incorporating actuator saturation and rate limits, and combining the proposed protocol with fast finite-time bipartite formation and obstacle-avoidance strategies for applications such as mobile robot or UAV swarms in cluttered environments.

CRediT authorship contribution statement

Tianhao Fei: Writing – original draft, Software, Methodology, Conceptualization; **Yongliang Yang:** Investigation, Writing – review & editing, Visualization, Validation; **Xiaowei Zhao:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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